

Mesh-to-Mesh Transfer Errors in BDF2 Unsteady Flow Simulations

Abstract

Discrete remeshing in an unsteady CFD calculation creates two coupled disturbances: the spatial discretization changes, and the histories required by a multistep integrator must be transferred to the new mesh. This study compares nearest-neighbor Taylor transfer, barycentric interpolation, and continuous Galerkin projection in BDF2 Euler and RANS calculations on fixed physical domains. Manufactured fields verify the nominal first- to third-order behavior and separate interpolation accuracy, boundedness, and integral preservation. For BDF2, the target-referenced history forcing is $2\varepsilon^n - \varepsilon^{n-1}/2$. A controlled Euler NACA 0012 restart shows why both histories matter: perturbing only the older state changes the lift even though the latest state is exact, and the measured response ratios agree with the BDF2 coefficients to within 0.3% in norm. Repeated Euler remeshing further shows that restoring a global integral can leave a spatial transport error essentially unchanged. The fixed-airfoil RANS extension adds SA and SST turbulence histories and separates pressure and friction loads. Galerkin projection preserves the monitored turbulence integrals much more closely than second-order nearest-neighbor transfer, but the two methods produce similar aggregate drag errors over the short two-seed comparison. Near-wall turbulence discrepancies also develop on vertices whose coordinates never move. Together, the results show that transfer accuracy, multistep-history consistency, and global conservation are distinct requirements; none alone provides a universal ranking of remeshing methods.

1 Introduction

Unsteady Reynolds-averaged Navier–Stokes (RANS) calculations use mesh adaptation to follow evolving flow structures without maintaining fine resolution throughout the domain. When adaptation regenerates connectivity, the flow and turbulence histories must be transferred to the new discretization before time integration can continue. The resulting calculation combines spatial approximation, solution transfer, and temporal integration; improving one component does not necessarily reduce their combined error. Time-accurate anisotropic adaptation and space–time adaptation for implicit RANS calculations illustrate this coupling [1, 2].

Two questions arise at each mesh change. Does the transfer represent the source field accurately on the target mesh, and does the transferred state permit a consistent continuation of the target discrete evolution? Polynomial reproduction and integral conservation address the first question only partially. A transfer can preserve a total integral while redistributing the field in a way that changes a residual balance or an aerodynamic load. Conversely, an accurate pointwise interpolant can introduce a small global imbalance. Repeated transfers may accumulate either effect, depending on the error’s spatial structure and its interaction with the evolving flow.

The distinction between interpolation accuracy and preservation of a discrete state is well established. Common-refinement Galerkin projection provides an integral-preserving, best- L_2 transfer between finite-element spaces [3, 4], while constrained variants are needed when additional discrete properties must be preserved [5]. Balance-preserving transfer in shallow-water models provides a related example: an accurate interpolant need not preserve a steady discrete balance [6,

7]. Here these ideas are examined in the specific setting of a vertex-centered finite-volume BDF2 restart and its subsequent aerodynamic response.

The comparison uses three established constructions: nearest-neighbor Taylor transfer, barycentric interpolation on a simplicial representation of the source mesh, and continuous Galerkin projection. Their nominal orders range from one to three. A bounded global correction is also applied to the pointwise methods, allowing the effect of enforcing a total integral to be assessed without replacing the underlying interpolation rule. The analysis distinguishes the consistent finite-element integral from the nodal integral actually used by the receiving solver, particularly when a quadratic projected field is reduced to vertex-only output.

The numerical argument proceeds from static transfer verification to accumulated Euler transport and controlled aerodynamic restarts. A two-dimensional Riemann calculation tests transfer frequency and deformation amplitude. An Euler NACA 0012 case isolates the two-history response on independent meshes and examines repeated load disturbances. The final application retains the fixed airfoil but adds SA and SST turbulence transport and viscous loads. Its nominally steady baseline makes remeshing-induced transients observable without imposed body motion. The scope is discrete replacement of stationary meshes in time-accurate calculations, not continuous mesh motion, a physically periodic RANS response, or a comparison of turbulence-model accuracy.

2 Solution-transfer operators

2.1 Representation and reconstructed derivatives

The source and target meshes, A and B , cover the same discretized domain Ω . Their nodal values are U_A and U_B , and $M_M = \text{diag}(w_i^M) \otimes I$ contains the positive control-volume weights of the vertex-centered flow solver on mesh M . For one component ϕ , the solver’s integral is

$$Q_M(\phi_M) = \sum_i w_i^M \phi_{M,i}. \quad (1)$$

All compared transfers return values at the solver vertices. Density, momentum, and total energy are transferred in conserved-variable form; nonlinear products are not reconstructed from separately interpolated primitive variables. Nodal sampling and control-volume averaging are different representation operators, even when the same weights are used to monitor an integral.

Nearest-neighbor corrections and the quadratic source representation require derivatives recovered from surrounding nodal data. A weighted least-squares fit determines the coefficients in

$$\phi_j - \phi_i \simeq g_i \cdot r_{ij} + \frac{1}{2} r_{ij}^T H_i r_{ij}, \quad r_{ij} = x_j - x_i. \quad (2)$$

The degree-one fit omits the Hessian; the degree-two fit retains it. With a full-rank, suitably scaled stencil and smooth data, polynomial reproduction gives an $O(h)$ gradient error for the linear fit and $O(h^2)$ gradient and $O(h)$ Hessian errors for the quadratic fit. These are conditional estimates: irregularity, anisotropy, rank deficiency, and weighting can degrade derivative recovery [8, 9, 10, 11]. The third-order variants therefore use degree-two reconstruction. Their accuracy is checked numerically rather than inferred from the requested polynomial degree alone.

2.2 Nearest-neighbor and barycentric interpolation

Nearest-neighbor transfer selects the closest source vertex $n(j)$ to each target vertex and evaluates a Taylor expansion there. With $r_j = x_j^B - x_{n(j)}^A$, the three variants are

$$(\mathcal{I}_{AB}^{\text{NN1}} \phi_A)_j = \phi_{A,n(j)}, \quad (3)$$

$$(\mathcal{I}_{AB}^{\text{NN2}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j, \quad (4)$$

$$(\mathcal{I}_{AB}^{\text{NN3}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j + \frac{1}{2} r_j^T H_{n(j)} r_j. \quad (5)$$

Thus NN1 copies a value, NN2 accounts for local slope, and NN3 also accounts for curvature. No containing-element search is required. For $|r_j| = O(h)$ and accurate derivatives, their smooth-field errors are respectively $O(h)$, $O(h^2)$, and $O(h^3)$. The selected donor changes discontinuously across source-node Voronoi boundaries, so a smooth displacement need not produce a smooth change in the interpolation error.

Barycentric interpolation instead locates the source simplex containing the target point. A conforming triangulated or tetrahedralized view of the original mesh supplies this search geometry, using original vertices rather than an augmented control-volume point set. If the containing simplex has vertices $x_{i_r}^A$, its barycentric coordinates satisfy

$$x_j^B = \sum_{r=1}^{d+1} \lambda_r x_{i_r}^A, \quad \sum_r \lambda_r = 1, \quad \lambda_r \geq 0. \quad (6)$$

They express the target point as a geometric weighted average of the simplex vertices. Applying the same weights to the field gives

$$(\mathcal{I}_{AB}^{\text{BAR2}} \phi_A)_j = \sum_r \lambda_r \phi_{A,i_r}, \quad (7)$$

$$(\mathcal{I}_{AB}^{\text{BAR3}} \phi_A)_j = \sum_r \lambda_r \left[\phi_{A,i_r} + \frac{1}{2} g_{i_r} \cdot (x_j^B - x_{i_r}^A) \right]. \quad (8)$$

BAR2 is affine-exact and already second order. For a quadratic field with exact vertex gradients, the half-gradient term in BAR3 cancels the quadratic interpolation remainder, yielding third-order consistency. The factor $1/2$ follows from this cancellation rather than from calibration. Both expressions are continuous across a conforming simplex interface when shared vertices have unique values and gradients. Neither enforces Eq. (1) between meshes.

Limiting is separate from nominal order. A factor $0 \leq \Phi_i \leq 1$ may scale both derivative terms in NN3, so $\Phi_i = 0$ recovers NN1; in BAR3 it scales the gradient correction and recovers BAR2, not a piecewise-constant interpolant. The smooth verification uses unlimited reconstruction. The discontinuous tests compare unlimited and Barth–Jespersen-limited transfer [12]; differentiable and higher-order limiter constructions provide related alternatives [13, 14, 15]. Damping a quadratic with a factor chosen from linear increments alone does not establish boundedness at every evaluation point. The measured extrema and the final thermodynamic admissibility of the state remain necessary diagnostics.

2.3 Galerkin projection and common-refinement integration

Galerkin projection determines a coupled target field rather than evaluating each target point independently. Let V_B^p be the continuous degree- p Lagrange space on the target simplicial mesh. The projection of a source representation ϕ_A^h satisfies

$$(\phi_B^h - \phi_A^h, v_B)_{L_2(\Omega)} = 0 \quad \forall v_B \in V_B^p, \quad (9)$$

and minimizes the L_2 distance to that source representation [3, 4]. For $\phi_B^h = \sum_j v_j N_j^B$, this gives

$$H_B v = b, \quad (H_B)_{ij} = \int_{\Omega} N_i^B N_j^B dx, \quad b_i = \int_{\Omega} N_i^B \phi_A^h dx. \quad (10)$$

The consistent mass matrix H_B is sparse and positive definite. Its off-diagonal entries couple basis functions sharing target elements; solving the system determines all coefficients simultaneously. It is distinct from the diagonal finite-volume matrix M_B used in time integration.

The load vector is the cross-mesh operation. On one target element, N_i^B has a fixed polynomial expression, but the source field may cross several source-element boundaries and change polynomial expression within that element. Consequently,

$$b_i = \sum_{K_B} \sum_{K_A: |K_A \cap K_B| > 0} \int_{K_A \cap K_B} N_i^B \phi_A^h dx. \quad (11)$$

Each intersection fixes both polynomial expressions and permits exact polynomial quadrature. These intersections form the source–target common refinement; they may be integrated locally without storing a complete additional mesh. Coverage must be complete and nonoverlapping. Products require degree-two quadrature for P^1/P^1 transfer and degree-four quadrature for P^2/P^2 transfer on affine simplices [16, 17]. This geometric integration and the mass-matrix solve distinguish projection from the point searches in NN and BAR.

GP2 uses P^1 source and target spaces. GP3 uses P^2 spaces, introducing edge coefficients internally. With $d_{ij} = x_i - x_j$ and $\bar{H}_{ij} = (H_i + H_j)/2$, the source midpoint value is reconstructed as

$$\phi_{ij} = \frac{1}{2}(\phi_i + \phi_j) - \frac{1}{8} d_{ij}^T \bar{H}_{ij} d_{ij}. \quad (12)$$

For a quadratic, the subtracted curvature term is exactly the difference between the endpoint average and the midpoint value. GP3 then solves for the complete quadratic target field, but only its original-vertex values are supplied to the flow solver. Nominal L_2 approximation orders are two and three under the usual smoothness and mesh-stability assumptions; nodal accuracy after this complete pipeline is assessed separately.

On mixed-element grids, BAR and GP use conforming subdivisions based on a deterministic shared-face diagonal rule. The approximation is P^p on the simplices, not Q^p on the native quadrilaterals or hexahedra. A valid conforming split and the chosen control-volume weights must therefore be retained consistently in geometric and integral checks.

2.4 Which integral is conserved?

For an unconstrained target space containing constants, Eq. (9) preserves the integral of the complete finite-element field. Coordinate functions also belong to continuous P^1 , and quadratic coordinates belong to P^2 . Exact projection therefore preserves the corresponding consistent moments before constraints, sampling, or limiting. This is not finite-volume conservation over arbitrary target cells: their indicator functions generally do not belong to the continuous test space. Nor is the inverse mass matrix a local transfer operator. Equality of source and target geometry on one subregion does not by itself make the projection the identity there.

For GP2, the consistent and solver-lumped integrals coincide when $w_i^M = \int N_i^M dx$ on both meshes. For GP3, source lifting and target vertex restriction each introduce a representation-dependent difference. In particular, a third-order projected field does not imply an $O(h^3)$ lumped-integral defect. Appendix E.1 gives a quadratic counterexample with an $O(h^2)$ discrepancy. The

consistent projection, the stored restart field, and any repaired field are consequently distinguished throughout the comparison.

A global correction can enforce a chosen solver integral independently of the transfer construction. For a raw nodal field ϕ_B^r , it seeks a correction c such that

$$\sum_j w_j^B c_j = Q_A(\phi_A) - Q_B(\phi_B^r), \quad \ell_j \leq \phi_{B,j}^r + c_j \leq u_j. \quad (13)$$

The implemented repair redistributes the defect over nodes with available capacity, freezing saturated nodes [18, 19]. Feasibility requires the requested integral to lie between the weighted lower and upper bounds, with fixed boundary values included. Source-range bounds on individual conserved components do not alone guarantee positive pressure. Accuracy is retained only if the correction is sufficiently small in the accuracy norm; a small total defect does not itself bound every nodal increment.

The optional NN/BAR correction is denoted by the suffix GC. GP additionally uses a projection-level limiter: source-vertex values in overlapping elements define $[\ell_j, u_j]$, widened by τ times the global source range. Clipping is followed by redistribution preserving a selected lumped total. Hard limiting sets $\tau = 0$; soft limiting allows a controlled excursion. Once active, this nonlinear post-processing need not preserve Galerkin orthogonality, first moments, or the best-approximation property. For GP2, preserving its lumped total preserves its consistent mass only under the weight compatibility above. For vertex-only GP3, no such implication holds automatically. These qualifications matter when interpreting the method labels in the numerical results.

3 BDF2 response to a mesh change

3.1 Reference trajectories and history mismatch

Between remeshing events, the mesh is fixed and the semi-discrete equations are

$$M_M \dot{U}_M + F_M(U_M, t) = 0. \quad (14)$$

The residual can contain Euler, viscous, or RANS terms; the state must include every transported variable required by the time integrator. The constant-step BDF2 equation on the target mesh is

$$\Phi_B(Y; V, W) = \frac{M_B}{\Delta t} \left(\frac{3}{2}Y - 2V + \frac{1}{2}W \right) + F_B(Y, t^{n+1}) = 0. \quad (15)$$

This form is not a general moving-volume ALE formula. Continuously moving meshes require consistent geometric history terms and a discrete geometric conservation law [20, 21]; those are outside the fixed-mesh-between-transfers derivation.

Let U_A^k and U_B^k be native trajectories for the same physical problem, computed on the two meshes. The transferred histories and their target-referenced discrepancies are

$$\hat{U}_B^k = \mathcal{I}_{AB}(U_A^k), \quad \varepsilon^k = \hat{U}_B^k - U_B^k, \quad k = n, n-1. \quad (16)$$

Here $\mathcal{I}_{AB}$ includes all limiting, correction, and restart-field processing. Let $\Pi_M u^k$ represent a common exact field in the solver's chosen data representation, and $e_M^k = U_M^k - \Pi_M u^k$. An exact decomposition, valid also for nonlinear transfer, is

$$\begin{aligned} \varepsilon^k = & \underbrace{\mathcal{I}_{AB}(\Pi_A u^k) - \Pi_B u^k}_{\text{transfer consistency}} \\ & + \underbrace{\mathcal{I}_{AB}(U_A^k) - \mathcal{I}_{AB}(\Pi_A u^k) - e_B^k}_{\text{propagated source error and target error}}. \end{aligned} \quad (17)$$

Only for a linear transfer may the second line be replaced by $\mathcal{I}_{AB}e_A^k - e_B^k$.

Thus a native-target reference does not make ε^k a pure interpolation error. It puts both states into the same target evolution problem while retaining their pre-existing discretization mismatch. For an output G , the decomposition

$$\begin{aligned} G_B(\hat{U}_B^k) - G_A(U_A^k) &= [G_B(U_B^k) - G_A(U_A^k)] \\ &\quad + [G_B(\hat{U}_B^k) - G_B(U_B^k)] \end{aligned} \quad (18)$$

separates native mesh sensitivity from the target-referenced restart discrepancy, not from an independently known consistency error. Comparisons of different transfers from the same source histories to the same target provide the controlled method comparison.

3.2 Exact history forcing and local state response

At any common trial state Y , changing only the histories gives the exact identity

$$\Phi_B(Y; U_B^n + \varepsilon^n, U_B^{n-1} + \varepsilon^{n-1}) - \Phi_B(Y; U_B^n, U_B^{n-1}) = -\frac{M_B}{\Delta t} \eta^n, \quad \eta^n = 2\varepsilon^n - \frac{1}{2}\varepsilon^{n-1}. \quad (19)$$

This identity does not require a linear spatial residual or small transfer errors. It tests history assembly provided both residual evaluations use exactly the same trial state and residual convention.

For the solved states, define $\delta^{n+1} = \hat{U}_B^{n+1} - U_B^{n+1}$. Let r and $\hat{r}$ be the remaining BDF2 algebraic residuals. Subtraction and expansion about the native target state give

$$A_B \delta^{n+1} = M_B \eta^n + \Delta t(\hat{r} - r) - \Delta t \mathcal{N}_B(\delta^{n+1}), \quad A_B = \frac{3}{2}M_B + \Delta t J_B, \quad (20)$$

where $J_B = \partial F_B / \partial U$ and $\mathcal{N}_B$ is the nonlinear Taylor remainder. With a locally Lipschitz Jacobian, $\mathcal{N}_B(\delta) = O(\|\delta\|^2)$. If A_B is locally invertible, the perturbations are small, and algebraic errors are negligible,

$$\delta^{n+1} = R_B \eta^n + O(\Delta t \|\eta^n\|^2), \quad R_B = A_B^{-1} M_B. \quad (21)$$

The constants are local to the mesh, state, time step, and norm; they are not assumed uniform over arbitrary anisotropic mesh families. Active limiter switches or insufficient inner convergence may invalidate this smooth-response approximation, but not Eq. (19).

The steady experiment uses one spatial discrepancy $\varepsilon^* = \mathcal{I}_{AB}U_A^* - U_B^*$ and assigns it independently to the two histories:

History configuration	ε^n	ε^{n-1}	η^n
Native	0	0	0
Full	ε^*	ε^*	$3\varepsilon^*/2$
Latest only	ε^*	0	$2\varepsilon^*$
Older only	0	ε^*	$-\varepsilon^*/2$

These diagnostic restarts predict

$$\delta_n \simeq \frac{4}{3}\delta_{\text{full}}, \quad \delta_{n-1} \simeq -\frac{1}{3}\delta_{\text{full}}, \quad \delta_{\text{full}} \simeq \delta_n + \delta_{n-1}. \quad (22)$$

The older-only case distinguishes the two-history forcing from a latest-state diagnostic. All three perturbations nevertheless use the same spatial direction: recovering their ratios is a test of directional response, not a complete validation of R_B for arbitrary history fields.

3.3 Aerodynamic response and propagation

For a differentiable aerodynamic functional with gradient g_B^{n+1} , the first-step discrepancy is

$$\Delta G^{n+1} = (g_B^{n+1})^T R_B \eta^n + O(\|\eta^n\|^2). \quad (23)$$

Equivalently, $A_B^T z_B = g_B^{n+1}$ gives $\Delta G^{n+1} = z_B^T M_B \eta^n$ to first order [22]. The error is weighted by the target dynamics and the output sensitivity, not by a constant conservation test. Conservation of the source integral constrains only

$$\mathbf{1}^T M_B \varepsilon^k = Q_A(U_A^k) - Q_B(U_B^k), \quad (24)$$

which need not vanish because the native integrals may differ.

A slope discrepancy must include the output difference already present at the transfer time:

$$\frac{\Delta G^{n+1} - \Delta G^n}{\Delta t} \simeq \frac{(g_B^{n+1})^T R_B \eta^n - (g_B^n)^T \varepsilon^n}{\Delta t}. \quad (25)$$

Dividing the first-step output difference alone by Δt is generally not this quantity. Cancellation between the two terms also prevents a universal force-kink scaling from being inferred from interpolation order alone.

For later steps, linearization along the native target trajectory gives

$$\begin{bmatrix} \delta^{k+1} \\ \delta^k \end{bmatrix} \simeq \begin{bmatrix} 2R_B^{k+1} & -R_B^{k+1}/2 \\ I & 0 \end{bmatrix} \begin{bmatrix} \delta^k \\ \delta^{k-1} \end{bmatrix}. \quad (26)$$

Repeated application connects the initial histories to a delayed output. This explains why a first-step force difference need not bound the later excursion. A dominant disturbance originating away from the airfoil may become measurable after acoustic or convective transport, but neither the inverse mass matrix of projection nor an implicit time step has compact support. Distance-dependent arrival is therefore measured by amplitude thresholds, not by a claim of strict first-nonzero causality.

4 Numerical experiments and assessment

4.1 Static transfer and repeated Euler transport

The static tests separate transfer error from flow-solver error. The asymmetric smooth density is

$$\phi = \rho_\infty \left[2 + 0.3e^{0.8x/L_x + 0.5y/L_y} \sin(2\pi x/L_x) \sin(2\pi y/L_y) \right]. \quad (27)$$

The field is transferred to structured quadrilateral and independently generated mixed triangular/quadrilateral target meshes at nominal resolutions 64, 128, and 256; the source mesh has half the target nominal resolution. Errors are relative nodal density L_2 differences from the analytic value at the target points. The effective spacing is defined as $h_{\text{eff}} = N_{\text{nodes}}^{-1/2}$; on this fixed domain it is used only to define refinement ratios. The discontinuous test replaces the smooth field with a circular inclusion of radius $0.25 \min(L_x, L_y)$ and density levels 1.2 and 3.6. Because the curved interface is not mesh-aligned, unlimited and limited transfers can be compared on a common unresolved discontinuity.

Consecutive and fitted orders are obtained from

$$p_{k,k+1} = \frac{\log(E_k/E_{k+1})}{\log(h_k/h_{k+1})}, \quad \log E \simeq p_{\text{fit}} \log h + b. \quad (28)$$

The three-level sequences are used to check consistency with the nominal orders, not to estimate mesh-independent asymptotic constants. A second smooth test on the structured Riemann mesh family keeps the mean displacement-to-spacing ratio near 0.057 and is reported in Appendix A.

The unsteady Riemann calculation solves two-dimensional Euler flow on $[0, 5] \times [0, 1]$, with a uniform 501×101 -node quadrilateral mesh, $h = 0.01$, $\Delta t = 10^{-3}$, and 400 physical steps. At each event, the original mesh receives a new bounded random displacement in x , with boundary vertices fixed. Distortion is not accumulated geometrically from one event to the next. The bound is the minimum incident corner-triangle altitude divided by SafeFactor, denoted SF. Two transfers per block give 80 transfers for a ten-step interval or 400 for a two-step interval. SF4 is the baseline; SF2 doubles the displacement bound. A no-remeshing calculation supplies a fixed-mesh reference. This experiment measures the combined effect of changing spatial discretization and transferring the solution, not a pure interpolation error relative to an exact flow.

Interpolation-only integral changes are sampled immediately before and after each transfer and summed separately from physical evolution. Density and energy are used because their reference integrals are away from zero. Spatial transport is assessed with two distinct diagnostics: maximum-gradient location and jump/maximum-gradient thickness, or the 50% crossing and 10–90% width of a sampled y -averaged density profile. Their reference positions, 2.6984 and 2.3993, are not interchangeable. The latter tracker is used for the frequency/amplitude comparisons. The transition is broad; these are operational profile diagnostics rather than independently validated sharp-shock positions. Only one random sequence is available, and the NN1 crossing is sensitive to the sampling grid.

4.2 NACA 0012: independent meshes and controlled histories

The aerodynamic case is Euler flow over NACA 0012 at $M_\infty = 0.1$ and $\alpha = 6^\circ$, with $p_\infty = \rho_\infty = T_\infty = 1$, $a_\infty = \sqrt{1.4} = 1.18322$, and $U_\infty = 0.118322$. BDF2 dual-time integration uses $\Delta t = 10^{-3}$. Forces are pressure-only, with unit reference chord and area and moment reference $x/c = 0.25$. The nonzero steady computed drag is treated as a numerical load, not as skin friction or shock-induced wave drag; absolute coefficient changes are used.

The independent source and target meshes contain 16,255/32,077 and 22,508/44,583 nodes/triangles, respectively. Both share the same 370 airfoil vertices and 63-node farfield polygon at 100 chords. The surface geometry is therefore unchanged although the interior meshes differ. Each native steady solution is advanced with equal BDF2 histories to create its own reference trajectory. Transfers from A to B then populate the history configurations of Eq. (22). All comparisons of their solved responses use the native target trajectory at the same physical step.

State discrepancies use the nondimensional volume-weighted norm $\|v\|_M = (v^T M v)^{1/2}$, summed over the stored flow components. Because the farfield occupies most of the domain, this norm emphasizes the global state and is complemented by near-wall and force diagnostics. Vector superposition is assessed by

$$e_{\text{sup}} = \frac{\|\delta_{\text{full}} - \delta_n - \delta_{n-1}\|_{M_B}}{\|\delta_{\text{full}}\|_{M_B}}, \quad (29)$$

with the analogous scalar expression for lift. Signed force ratios and positive state-norm ratios are kept distinct.

The solver exports $-\Phi_B$, including the temporal residual. Initial residuals are compared before the first update and only for cases sharing a trial state: older-only versus native at U_B^* , and latest-only versus full at $\hat{U}_B^*$. Their $\Delta\eta$ values are respectively $-\varepsilon^*/2$ and $+\varepsilon^*/2$. The sign is converted to the convention of Eq. (19). Single-precision exports limit the interpretation of the smallest residual

differences. First-step forces and subsequent 50-step summaries provide separate assessments of immediate and later response.

4.3 Repeated airfoil remeshing and localized disturbances

The repeated-remeshing experiment perturbs the source mesh between wall distances 0.01 and 20 chords while keeping the boundaries fixed. Thirty-nine mesh replacements occur, one every ten physical steps in the 400-step window. NN2 and GP2 are evaluated on the same two random deformation sequences, and force differences are formed relative to the no-remeshing trajectory. Least-squares slopes are quoted per 100 physical steps as descriptive summaries; the associated regression standard errors refer to each individual correlated time series and are not ensemble uncertainties.

A separate one-event experiment perturbs narrow wall-distance bands centered at $d/c = 0.05$, 0.10, and 0.20, then continues without further adaptation. Force onset is the first crossing of a selected fraction of that case’s peak response. Fixed pressure shells provide a less spatially integrated diagnostic through their root-mean-square pressure difference. A shell a distance $d - r$ inside the launch band has the approximate acoustic lag $(d - r)/(a_\infty \Delta t) \simeq 845(d - r)$. This is a low-Mach scale estimate; actual characteristics depend on local velocity, sound speed, and propagation direction.

The no-remeshing trajectory is the reference for this localized-event experiment, whereas the mixed-history study uses an independently evolved target-mesh reference. Even though the meshes coincide outside the displaced band, the transfer is not guaranteed to remain strictly local. An immediate response is observed up to about $0.08c$ inward of the band; those shells are excluded from the propagation fit. Projection coupling, limiter redistribution, the changed spatial operator, and the implicit update can all contribute to this near-instantaneous component.

4.4 Fixed-body RANS extension

The RANS extension uses the fixed NACA 0012 at $M_\infty = 0.1$, $\alpha = 6^\circ$, and $Re_c = 6 \times 10^6$, with Spalart–Allmaras (SA) and shear-stress-transport (SST) closures [23, 24]. The objective is to measure remeshing-induced transients about a nominally steady viscous solution, including turbulence histories and skin-friction loads. Both closures use the same hybrid mesh with 29,956 vertices, 14,070 quadrilaterals, 31,339 triangles, 370 airfoil vertices, and a farfield at $100c$. The first off-wall spacing is approximately $5 \times 10^{-6}c$, and the quadrilateral region extends to about $0.053c$. Surface values reach $y^+ \simeq 2.69$; the mesh is therefore treated as the resolution used for this comparison, not as an independently grid-converged wall treatment.

Mean-flow fluxes use HLLC with MUSCL reconstruction and a Venkatakrishnan limiter. Turbulence convection uses scalar upwinding without MUSCL reconstruction. The second-order transfer label therefore does not describe the order of every term in the receiving RANS discretization. BDF2 uses $\Delta t = 10^{-3}$, 50 inner iterations per physical step, CFL 6, and FGMRES with ILU preconditioning and a 60-iteration linear limit. Here the velocity reference is the freestream sound speed: $a_\infty = 1$ and $U_\infty = 0.1$, unlike the Euler normalization in Sec. 4.2. The 400-step comparison spans only $tU_\infty/c = 0.04$. Each closure has its own 800-step no-transfer reference, initialized with the same steady and startup histories as its perturbed calculations. References are subtracted at matching steps; their startup relaxation is not treated as statistical noise.

The repeated-remeshing comparison uses NN2 and limited GP2 with two prescribed deformation sequences and one mesh replacement every ten physical steps. Surface vertices and the 12,233 vertices with $d_w < 0.01c$ are fixed; nodes farther from the wall may move out to $20c$. SF4 sets the baseline displacement limit and a mesh-quality floor of 0.2 is enforced. For each seed, NN2 and

GP2 see exactly the same mesh sequence. Thirty-nine replacements are followed by at least one physical step in the analyzed window; the terminal history export is excluded because no subsequent force sample is available. A separately converged solution on the first displaced mesh quantifies the change in the native discrete equilibrium. The one-event GP2 cases perturb the bands $[0.05, 0.10]c$, $[0.10, 0.15]c$, and $[0.20, 0.25]c$, with selected SF2 and longer-window variants.

Density, momentum, energy, and both BDF2 histories of the turbulence variables are transferred. Conservation is monitored for density-weighted $\tilde{v}$ in SA and for density-weighted k and ω in SST. After transfer, SST enforces the primitive-variable floors $k \geq 10^{-10}$ and $\omega \geq 10^{-4}$; SA has no analogous post-pass. Wall distance and eddy viscosity are recomputed on the receiving mesh. The comparison therefore concerns the complete NN2 and GP2 restart pipelines, including limiting and positivity processing. It does not isolate turbulence-history error from the simultaneous mean-flow transfer or test a separate RANS global-correction variant.

Total loads are taken from the force histories. Pressure and friction contributions are additionally integrated from the exported surface fields using the same quadrature for case and reference. These differences resolve changes hidden by the six-decimal force-breakdown output, but the underlying surface fields remain single precision. Surface records duplicated at identical physical steps are counted once. The corresponding script, precision limitations, event accounting, and supporting diagnostics are documented in Appendix F.

4.5 Evidence and reproducibility limits

The convergence orders in the static tests can be recomputed directly from the stored error tables. The Riemann and Euler NACA datasets are mostly reduced summaries, while the RANS dataset also includes force histories, surface fields, and transfer-integral diagnostics. Raw restart states, complete residual vectors, case configurations, and full inner-iteration histories are not included. Numerical settings are therefore treated as run conditions, and conclusions that would require those missing quantities are kept outside the claims of the paper.

5 Results

5.1 Accuracy and boundedness of one-time transfer

Figure 1 shows the smooth-field refinement results. NN1 is first order; NN2, BAR2, and GP2 are consistent with second order; NN3 is consistent with third order. BAR3 and GP3 have fitted slopes above three over the three-level interval, approaching approximately 3.2 on the finest pair. This is finite-range convergence behavior, not evidence of a higher designed order. On the finest mixed grid, the relative errors are approximately 2.6×10^{-7} for BAR3, 3.4×10^{-7} for GP3, and 2.6×10^{-6} for NN3. These differences resolve smooth interpolation performance on the tested mesh family; they do not predict a corresponding ranking of restarted flow solutions.

The discontinuous test has a different regularity limit (Fig. 2). If an $O(1)$ error occupies an $O(h)$ neighborhood of the interface, its continuous L_2 norm scales as $h^{1/2}$. The fitted nodal-error slopes, approximately 0.51 on the structured grid and 0.43–0.45 on the mixed grid, are compatible with this mechanism. The individual intervals are less uniform; for example, the finest structured NN1 slope is approximately 0.33. The short sequence therefore supports reduced-order convergence, not a universal asymptotic rate for every variant. BAR3 and GP3 reduce the finest mixed-grid error from NN1’s 6.58×10^{-2} to approximately 4.92×10^{-2} and 5.02×10^{-2} , respectively.

Accuracy and boundedness do not give the same ranking. Unlimited NN2, NN3, and BAR3 overshoot the upper plateau by 0.800, 0.426, and 0.133, respectively. Barth–Jespersen limiting

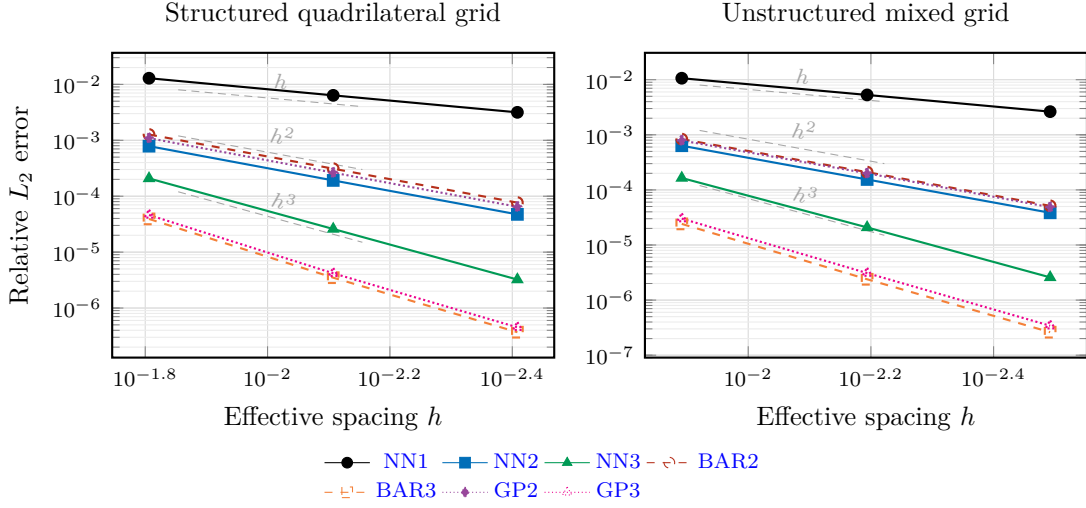


Figure 1: Convergence of the density-transfer error for the smooth asymmetric field. Curves show the relative nodal L_2 error as the mesh is refined; gray guides correspond to $O(h)$, $O(h^2)$, and $O(h^3)$.

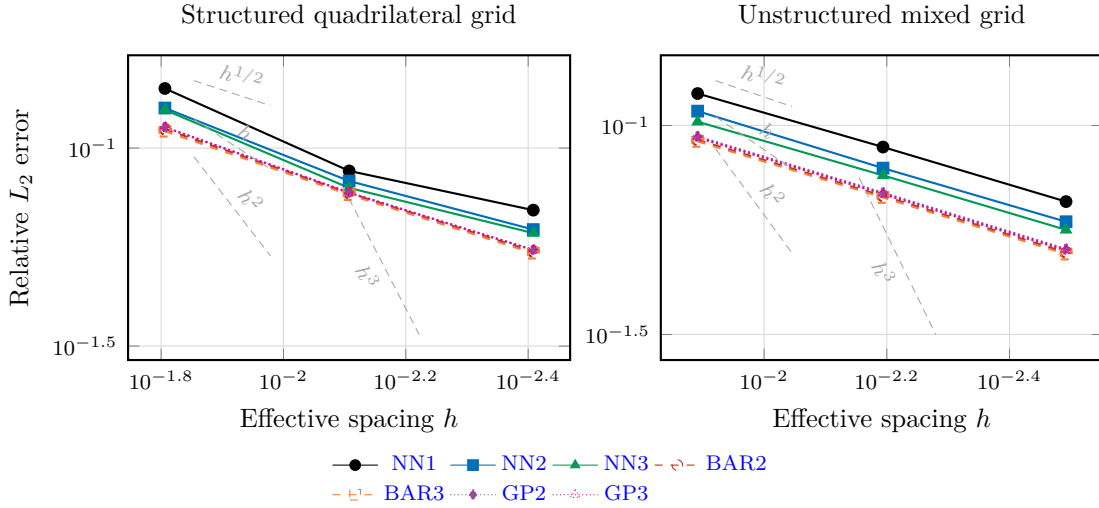


Figure 2: Density-transfer error for the discontinuous circular inclusion. The $O(h^{1/2})$ guide shows the expected scaling for a mesh-unresolved jump; the $O(h)$, $O(h^2)$, and $O(h^3)$ guides show how the smooth-field orders are lost at the discontinuity.

removes the measured over- and undershoots, and adding GC reduces the source-to-target integral defect to floating-point level. Hard-limited GP2 is also bounded with a similarly small defect in this test, whereas vertex-only GP3 remains bounded but has a lumped integral defect of about 3×10^{-4} . Appendix A gives the complete boundedness and conservation results.

A smooth single-transfer check on the Riemann mesh gives a complementary result: GC reduces the source-referenced density defect to approximately 10^{-15} for all seven base operators, while changing L_1 error by at most 0.64% at that resolution. This is a small measured correction cost, not a guarantee that global repair preserves the convergence order on every mesh family. Appendix A provides the full values and the independently verified L_1 refinement sequence.

5.2 Accumulated transport under repeated remeshing

After 80 transfers in the Riemann case, NN1 changes the density and energy integrals by +0.0203% and +0.0211% of their initial values. The remaining methods stay within 0.0023% in net density change, and GP2 has the smallest median absolute per-transfer density defect, 1.07×10^{-7} compared with 1.11×10^{-5} for NN1. Signed accumulation can contain cancellation, so it should not be read as a statistical drift rate. Boundedness gives a different picture: NN1 and BAR2 introduce no new density extrema, whereas the unlimited derivative-based variants overshoot at every sampled transfer. The complete accumulation and profile results are in Appendix B.

Figure 3 compares the final transition position and width as the remeshing becomes more frequent and more severe. At SF4, increasing the number of transfers from 80 to 400 raises the position error from 1.28 to 7.56 cells for NN1 and from 1.10 to 8.12 cells for BAR2. With 400 transfers and the larger SF2 deformation, the corresponding errors are 25.78 and 26.43 cells, whereas GP2 remains at 0.09 cell in the tabulated extraction. The absolute NN1 crossing is sensitive to the profile-sampling procedure, so these values establish the qualitative amplification seen in this extraction rather than a universal rate.

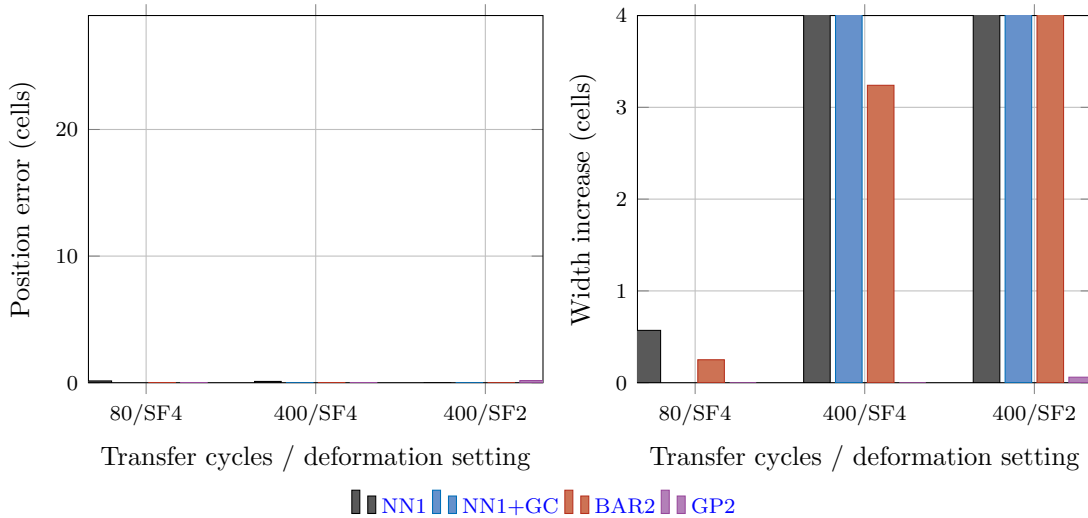


Figure 3: Effect of remeshing frequency and deformation amplitude on the density transition. The left panel shows the final position error and the right panel the increase in 10–90% width, both in baseline-cell units. SF4 is the baseline deformation and SF2 doubles the displacement bound; NN1+GC was not run for the 80-transfer case.

The raw/GC pair isolates the effect of enforcing the total density integral. At 400 transfers and

SF2, the cumulative integral change is reduced from 3.2×10^{-3} to 2.4×10^{-13} , yet the transition-position error changes only from 25.78 to 25.53 cells and the width increase from 3.53 to 3.25 cells. Global conservation therefore removes the scalar integral defect without removing the spatial transport error. This does not imply that finite-volume local conservation is necessary: higher-order pointwise methods also maintain small position errors in the frequency study, and the continuous Galerkin projection does not enforce arbitrary cellwise balances.

The moment and limiter diagnostics in Appendix C offer a possible interpretation, but not a causal identification. GP2 has smaller measured lumped centroid changes than BAR2 or NN1, consistent with the moment constraints of the unmodified projection. A whole-domain density centroid is not the 50%-crossing position, however. Similarly, the unlimited GP2 excursion rises from 7.6×10^{-6} to 3.6×10^{-3} between the two displacement settings, suggesting stronger limiter activity. A global excursion alone does not determine activation of every local bound, and no resolved limiter-on/off width budget is available. The observed width increase cannot therefore be partitioned between projection, limiter repair, spatial discretization, and profile sampling.

5.3 Target-referenced NACA restart response

The native airfoil coefficients differ before any transfer (Table 1). This baseline is essential: a restarted target state cannot be expected to reproduce the source-mesh force simply because the body geometry is shared. The native target trajectory also relaxes over its 51-step observation window, spanning 1.9×10^{-4} in lift, 1.4×10^{-4} in drag, and 5.7×10^{-5} in moment. Responses are differenced from that trajectory step by step. The span is a deterministic reference variation, not an independently established noise floor or a bound on inner-solve error.

Table 1: Steady Euler NACA 0012 solutions obtained independently on the source mesh A and target mesh B . Their load difference is the native mesh-to-mesh discretization difference against which the restart response is interpreted.

Mesh	Nodes	Triangles	C_L	C_D	C_m
A	16,255	32,077	0.69908019	0.00995221	0.00965742
B	22,508	44,583	0.70900224	0.00305722	0.00722284

For the transferred steady states, $\|U_B^*\|_{M_B} = 478.46$. The second- and third-order discrepancy norms range from 7.040×10^{-4} to 7.243×10^{-4} , a spread below 3%; NN1 gives 1.474×10^{-3} . This clustering is consistent with a shared native mesh mismatch, but does not separate that mismatch quantitatively from transfer error. The lowest tabulated norm belongs to BAR2, but the small spread does not establish a general ranking. Near-wall discrepancies can remain important despite relative whole-domain norms near 1.5×10^{-6} , because the normalization includes the large farfield.

The common-trial residual pairs satisfy Eq. (19) with relative vector discrepancies of 2.4×10^{-8} – 4.2×10^{-8} for NN/BAR, 4.4×10^{-6} for GP2, and 1.3×10^{-6} for GP3. This confirms the implemented BDF2 history signs and coefficients at the precision of the exported fields. The check concerns residual assembly; it does not test the accuracy of the subsequently converged nonlinear response.

Figure 4 provides the direct physical consequence of the older history. With $\varepsilon^n = 0$, the older-only lift difference is still 1.83×10^{-3} – 2.55×10^{-3} for the five methods with complete first-step force summaries, and its sign is opposite to the corresponding full-history response. These magnitudes exceed the native target window’s lift span, but that comparison is descriptive rather than a significance test. More importantly, a diagnostic based only on the latest history would assign zero perturbation to these cases, whereas the BDF2 forcing does not.

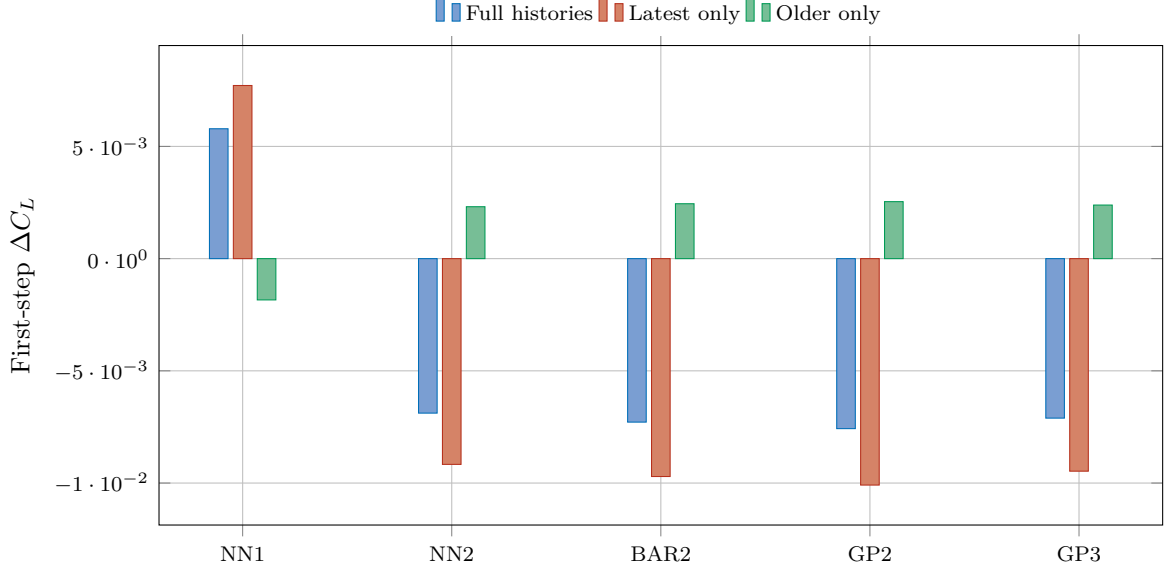


Figure 4: First-step lift change produced by the four BDF2 history configurations, measured relative to the native target-mesh continuation. In the older-only restart the latest state is unchanged ($\varepsilon^n = 0$), so its nonzero response isolates the contribution of the $n - 1$ history.

Table 2 tests the directional response more closely. The latest/full state-norm ratios lie between 1.3330 and 1.3333, and the older/full magnitudes between 0.3335 and 0.3341. Their agreement with $4/3$ and $1/3$ is within 0.3%. Signed lift ratios are within 1% of the corresponding predictions except NN1’s older/full ratio, which differs by approximately 4.7%. Vector superposition is less exact than the norm ratios: its defect is 1.8–3.5%, compared with 0.3–1.5% for lift. This is evidence for approximate directional linearity of the tested histories, not verification of a quadratic remainder law or of the full Jacobian/adjoint prediction. Neither an amplitude-convergence study nor a direct solve with A_B is available.

Table 2: Response ratios for the mixed-history restart. State ratios use positive norms, whereas the lift ratios retain their sign. The linearized BDF2 model predicts latest/full = $4/3$ and older/full = $1/3$ in norm, with a signed lift ratio of $-1/3$ for the older-only case.

Method	State: latest/full	State: older/full	Lift: latest/full	Lift: older/full
NN1	1.3333	0.3335	1.3329	−0.3179
NN2	1.3331	0.3341	1.3318	−0.336
NN3	1.3331	0.334	1.3309	−0.3362
BAR2	1.3332	0.334	1.3328	−0.3358
BAR3	1.3332	0.334	1.3329	−0.3357
GP2	1.333	0.334	1.3322	−0.3352
GP3	1.3331	0.334	1.3326	−0.3359

Global correction provides a second controlled test. The corrected target-referenced density difference is -8.12114×10^{-4} for both NN1 and BAR2, as expected from Eq. (24); it is not a residual source-conservation failure. For NN1, GC changes the state-discrepancy norm by -0.0109% , the full-history first-step norm by -0.00890% , and the lift response by $+0.0886\%$. For BAR2, the lift change is -0.00274% . The response is therefore weakly affected by the correction in these pairs.

The complete raw/corrected values are in Appendix D.

First-step agreement does not bound the later load excursion. The 50-step summaries show an NN1 peak lift difference of 2.697×10^{-2} , approximately 4.68 times its first-step value, followed by a final value of -1.390×10^{-4} . The listed higher-order cases finish with 22–39% of their first-step magnitudes. These endpoint and peak data support transient amplification for NN1 and subsequent reduction in the other cases; without the complete exported histories they do not independently establish monotonic decay at every intermediate step. Maximum full-history surface- C_p differences of 0.29–0.45 further show that a small global norm need not imply a small local pressure response.

5.4 Load accumulation and delayed aerodynamic response

The repeated-remeshing Euler airfoil case reveals behavior that a one-step norm cannot capture. Both deformation sequences produce a positive numerical-drag trend for GP2 and NN2 (Fig. 5); averaging the two fitted slopes gives 6.6×10^{-6} and 3.85×10^{-5} per 100 physical steps, respectively, a ratio of 5.83. GP2 therefore reduces, but does not eliminate, the observed accumulation. With only two sequences, these slopes demonstrate repeatability for the tested realizations rather than a population drift rate.

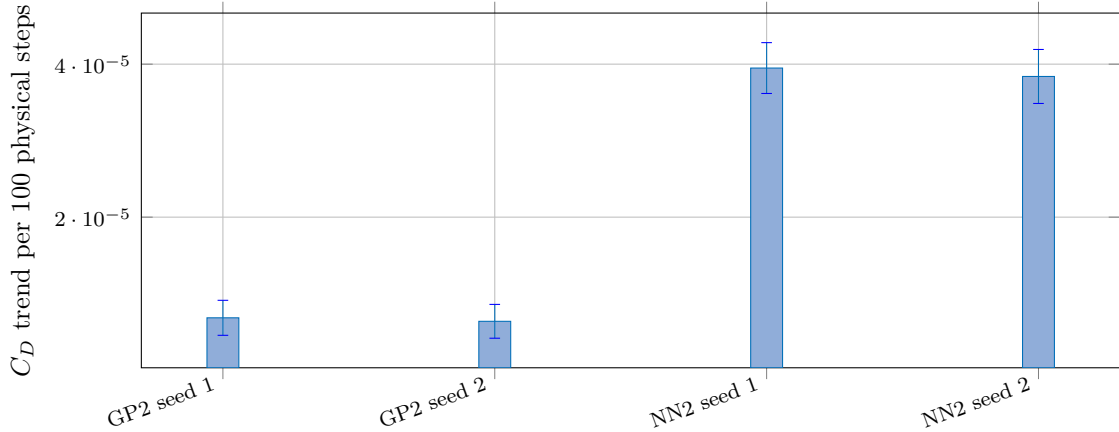


Figure 5: Mean numerical-drag trend over 39 repeated mesh replacements for GP2 and NN2, shown for two independent deformation sequences. Bars give the fitted change in C_D per 100 physical steps; whiskers are ordinary regression standard errors for each individual time series.

The other loads constrain a symmetry-based interpretation. GP2’s lift trend changes sign across seeds, whereas both NN2 lift slopes are positive and differ substantially in magnitude. NN2 also has closely matching negative moment trends, -1.46×10^{-5} and -1.49×10^{-5} per 100 steps. It is therefore not supported that lift and moment errors are generally zero-mean or that all non-drag effects average away. The drag comparison is compatible with accumulated numerical smoothing, but no entropy budget separates interpolation from the changed flow discretization. Conservation of a scalar integral alone would not imply absence of such a trend, consistent with the limitations of unconstrained projection discussed by Maddison and Hiester [5].

For a single localized remeshing event, force-onset delays increase with launch distance (Fig. 6). The 25% and 50% detectors bracket the approximate acoustic lag $845d$. Fits through the three radii have negative intercepts of roughly 8.5–18 steps in magnitude; these are not zero compared with the shortest delays. The high three-point R^2 values therefore do not establish a zero-intercept propagation law or uniquely exclude alternative onset mechanisms. Doubling displacement changes

the measured arrival by at most two steps and the peak by a factor 1.63. This demonstrates limited sensitivity to that amplitude change, but a fraction-of-own-peak detector is already insensitive to pure amplitude scaling and cannot alone prove causality.

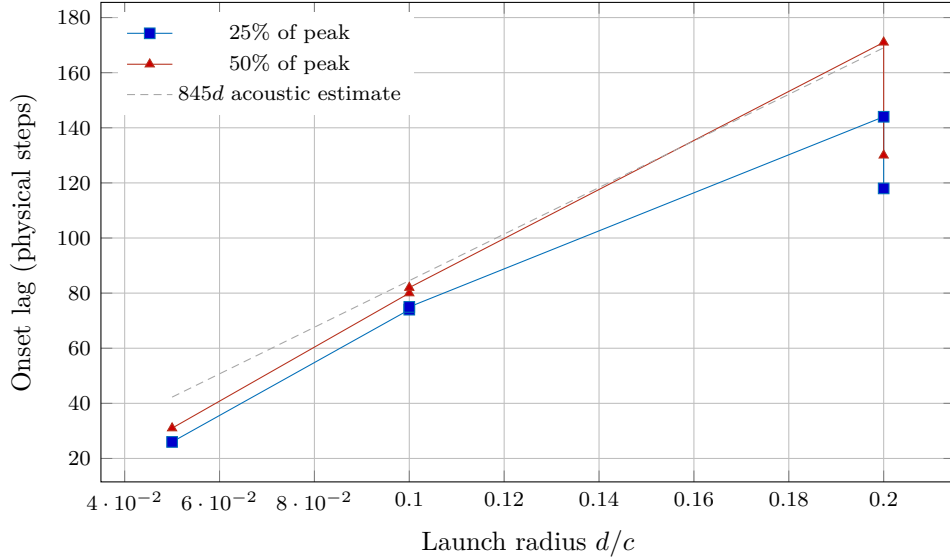


Figure 6: Delay between a localized mesh-replacement event and the subsequent lift response. Onset is defined at 25% and 50% of each case’s peak response and is plotted against the launch distance d/c ; the gray line is the freestream acoustic travel-time estimate $845d$.

The pressure-shell results give more direct support for acoustic transport. Across six thresholds applied to the same event, lag slopes range from 782 to 917 steps per chord, corresponding to speeds of 1.09–1.28. Their mean speed is approximately 1.184, and the threshold-to-threshold standard deviation is 0.079; this spread is a detector-sensitivity measure, not a confidence interval from independent realizations. The range contains $a_\infty = 1.18322$ and is well separated from the freestream convection speed 0.118322. A target-referenced check at the higher thresholds gives approximately 1.26–1.28, also on the acoustic scale. Together these observations support an acoustic interpretation of the dominant delayed disturbance while allowing an immediate, small nonlocal component. They do not verify the propagator or output adjoint quantitatively.

5.5 RANS load response on discretely changing meshes

The RANS cases give a different method comparison from the Euler runs. Figure 7 shows total- and friction-drag differences relative to the no-remeshing trajectory for each closure. Across the eight runs, the RMS total-drag difference is $(3.01\text{--}3.43) \times 10^{-5}$. For each deformation sequence, NN2 and GP2 differ by less than 1% in this aggregate measure, although their step-by-step difference still has an RMS of $(8.71\text{--}8.84) \times 10^{-6}$. No consistent method ordering emerges. The clear GP2 advantage in the Euler drag trend therefore does not carry over to this short RANS comparison.

The friction contribution decreases in every run, with fitted slopes of $-(2.86\text{--}4.52) \times 10^{-7}$ per 100 steps, while the larger total-drag excursions are dominated by pressure. The total-drag slopes are small compared with the oscillations, and the detrended histories have lag-one correlations of about 0.79–0.84. Appendix F therefore reports the fits only as descriptive summaries. With two deformation sequences over this short window, the data support a paired finite-time comparison, not a long-time drift law or a statistical equivalence claim.

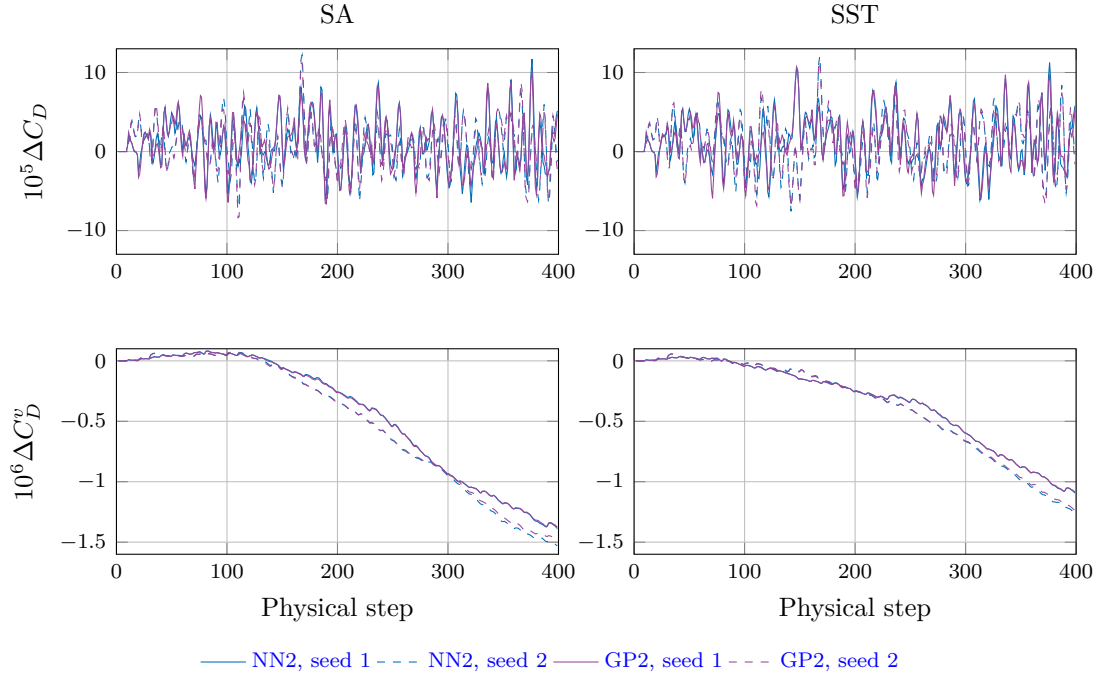


Figure 7: Drag response to repeated mesh replacement in the SA and SST calculations. The upper panels show the total-drag difference from the corresponding no-remeshing reference; the lower panels show the friction-drag contribution obtained from the surface fields. NN2 and GP2 use the same two deformation sequences within each closure. Note the different vertical scales in the two rows.

Table 3: Largest instantaneous change in each monitored integral when a RANS history state is transferred from the old mesh to the new one. Values are normalized by the source integral and pool both BDF2 histories, both deformation sequences, and the 39 mesh replacements followed by physical evolution (156 transfers per entry).

Closure	Inventory density	NN2	GP2
SA	ρ	$5.5 \cdot 10^{-11}$	$2.19 \cdot 10^{-13}$
SA	ρE	$4.26 \cdot 10^{-11}$	$2.87 \cdot 10^{-13}$
SA	$\rho \tilde{\nu}$	$2.48 \cdot 10^{-6}$	$4.08 \cdot 10^{-9}$
SST	ρ	$4.94 \cdot 10^{-11}$	$2.27 \cdot 10^{-13}$
SST	ρE	$4.16 \cdot 10^{-11}$	$3.03 \cdot 10^{-13}$
SST	ρk	$2.98 \cdot 10^{-6}$	$2.59 \cdot 10^{-8}$
SST	$\rho \omega$	$8.15 \cdot 10^{-8}$	$1.08 \cdot 10^{-9}$

Changing the mesh can itself alter the steady load. On the first displaced grid, the independently converged lift differs from the original-grid steady value by -1.891×10^{-3} for SA and -9.649×10^{-4} for SST. The corresponding mean lift deviations during the first ten-step GP2 continuation are instead $+1.318 \times 10^{-5}$ and $+1.150 \times 10^{-5}$. These are different questions: the first comparison measures native discrete equilibria, whereas the second measures a short continuation from transferred histories. The equilibrium difference cannot be subtracted as an instantaneous interpolation correction, nor do similar NN2 and GP2 loads show that changing the discretization is harmless. This is the distinction in Eq. (18) applied to the viscous case.

5.6 Turbulence transfer, integral preservation, and response localization

Table 3 compares the integral change introduced at the transfer itself. For the SA working variable, GP2 lowers the maximum relative defect from 2.48×10^{-6} to 4.08×10^{-9} . For SST, the corresponding reductions are 2.98×10^{-6} to 2.59×10^{-8} for k and 8.15×10^{-8} to 1.08×10^{-9} for ω ; density and energy show the same ordering. These are source-to-target transfer defects, not differences from a native target solution. They also do not bound local errors, particularly for density and energy integrated over a 100c domain. Once time integration resumes, turbulence production, destruction, and transport legitimately change these integrals.

This large difference in integral preservation is not mirrored by the aerodynamic loads in Fig. 7. It also does not keep the protected near-wall solution unchanged. At step 395, the eddy-viscosity difference on the fixed $d_w < 0.01c$ vertices has an RMS equal to 0.38–0.44% of the maximum reference value on that band, with similar values for NN2 and GP2. Because the coordinates of these vertices are identical, the comparison is direct; however, it does not separate the initial transfer perturbation from the response to the changed spatial operator and subsequent transport.

SST positivity enforcement is active during the repeated-remeshing runs. The largest per-transfer count is 415–423 vertices for GP2 and 13–19 for NN2. These counts do not measure the size of the correction, but they show that the field advanced by the solver is not the unconstrained projection alone. Appendix F gives the clipping and fixed-band diagnostics. Because the mean-flow and turbulence histories were transferred together, the present data do not isolate the contribution of the turbulence variables to the viscous-load response.

The localized RANS events do not resolve the propagation law observed in the Euler case. The 50%-of-peak lift onsets are 3, 8, and 5 steps for launch bands at increasing distance, while the two nearer cases reach their peak near the end of the observation window. Moreover, 44 of 72 pressure-shell peaks are smaller than three single-precision pressure increments, and many of the

remaining signals cross their thresholds within zero or one saved step. These records are therefore insufficient to determine a propagation speed. They show only that an immediate component is present at the available precision; they neither identify its mechanism nor rule out a later acoustic response.

6 Discussion and conclusions

The evidence distinguishes three requirements of a remeshed time-accurate calculation: adequate spatial transfer, appropriate multistep histories, and sufficiently accurate evolution on the receiving mesh. Global integral preservation addresses only part of the first requirement. Its insufficiency is demonstrated by paired Riemann raw/GC comparisons; the importance of the older history is isolated by the Euler NACA restart. The exact common-trial residual identity and approximate BDF2 coefficient ratios support the history analysis without constituting a full validation of the nonlinear resolvent or aerodynamic adjoint.

The fixed-body RANS cases add the two ingredients absent from the Euler tests: transported turbulence histories and viscous loads. GP2 preserves the monitored turbulence integrals much more closely than NN2, yet their total-drag RMS errors differ by less than 1% for each paired deformation sequence and show no consistent ordering. Both methods also produce a small reduction in friction drag and comparable eddy-viscosity changes inside the protected near-wall vertex set. Integral preservation is therefore a useful transfer diagnostic, but it does not by itself rank the aerodynamic response. The behavior of the complete post-processed restart state is what matters to the receiving RANS calculation.

Reference choice remains part of that evaluation. Independently converged grids have different discrete equilibria, and comparison with a no-remeshing trajectory includes changes in the spatial operator as well as transfer. Identical mesh sequences provide a controlled method comparison, not a pure interpolation-error measurement. Likewise, a spatially restricted mesh change need not produce a strictly localized field perturbation. The Euler pressure records support a dominant acoustic delay, whereas the RANS pressure precision, early nonlocal response, and window-dependent peaks do not establish a propagation law.

The conclusions are limited to fixed physical domains and short remeshing-induced transients about nominally steady airfoil flows. The RANS extension is sufficient to assess turbulence-history transfer and viscous loads within that scope; body motion is a separate problem. The present 50-inner-iteration budget, short time horizon, protected near-wall region, and two deformation sequences are not sufficient for a long-time accuracy ranking or a production recommendation. Those stronger claims would require targeted physical-step and time-step checks, complete run configurations, and transfer-cost measurements. Continuous moving meshes, separated periodic flows, and a higher-order RANS method ranking are left outside the scope of this paper.

A Additional static-transfer verification

The fitted orders supplement the trends in Section 5.1; the underlying generated files also retain consecutive-level estimates. Smooth orders above the nominal value are interpreted over the available finite refinement range. For the discontinuity, the $h^{1/2}$ guide is conditional on an $O(h)$ interface-error band, not a theorem about every exported nodal norm.

The conservation plots use the exported solver-lumped defects. These curves show the magnitude of the exported defect statistic. An independent consistent-integral check is required before a nonzero lumped discrepancy is assigned solely to projection or to vertex restriction. On the finest structured

Table 4: Observed density L_2 convergence orders for the smooth asymmetric field. The table reports the least-squares slope through all three mesh levels; the finest-grid pairwise slope is retained in the data files for reference.

Scheme	Nominal	Structured	Mixed
NN1	1	1.01	1.01
NN2	2	2.03	2.03
NN3	3	3	3
BAR2	2	2.02	2.02
BAR3	3	3.36	3.27
GP2	2	2.04	2.02
GP3	3	3.34	3.24

Table 5: Observed density L_2 convergence rates for the discontinuous circular inclusion. The $1/2$ reference rate corresponds to the error contributed by a mesh-unresolved interface.

Scheme	Band scaling	Structured	Mixed
NN1	0.5	0.51	0.43
NN2	0.5	0.51	0.44
NN3	0.5	0.52	0.43
BAR2	0.5	0.51	0.45
BAR3	0.5	0.51	0.45
GP2	0.5	0.52	0.45
GP3	0.5	0.51	0.45

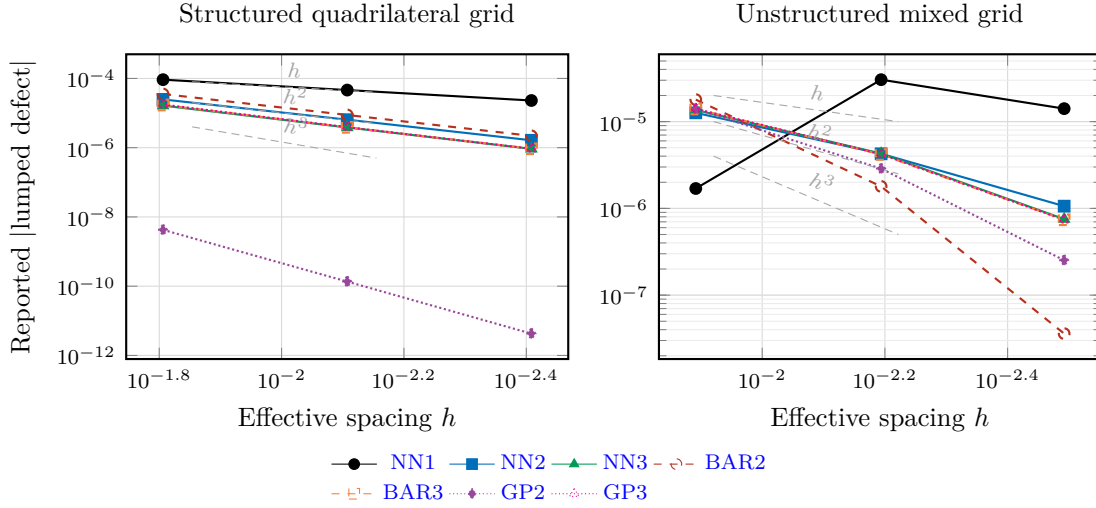


Figure 8: Convergence of the density-integral defect for the smooth asymmetric field before global correction. The ordinate is the absolute difference between source and target solver-lumped integrals; gray guides show $O(h)$, $O(h^2)$, and $O(h^3)$ slopes.

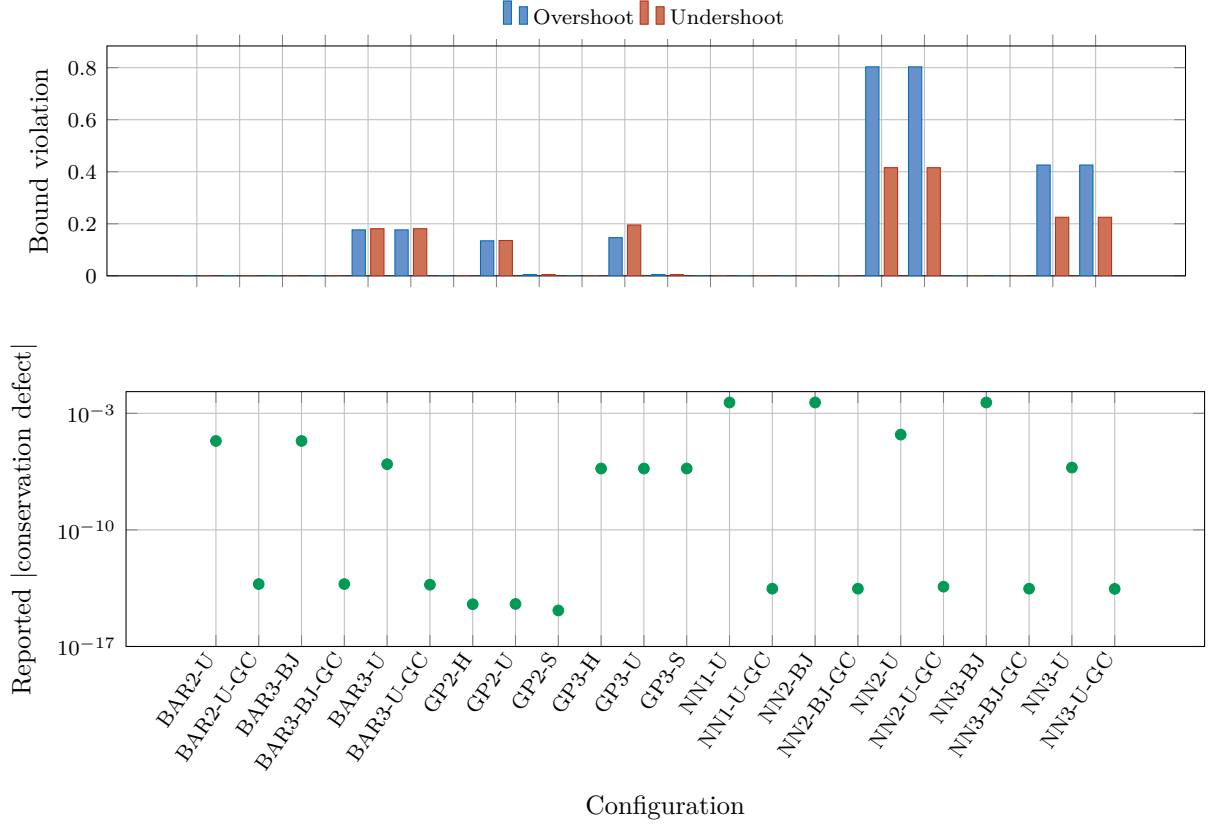


Figure 9: Boundedness and solver-lumped conservation for the discontinuous circular inclusion. U denotes unlimited reconstruction, BJ the Barth–Jespersen limiter, GC the bounded global correction, and S/H the soft and hard Galerkin limiters.

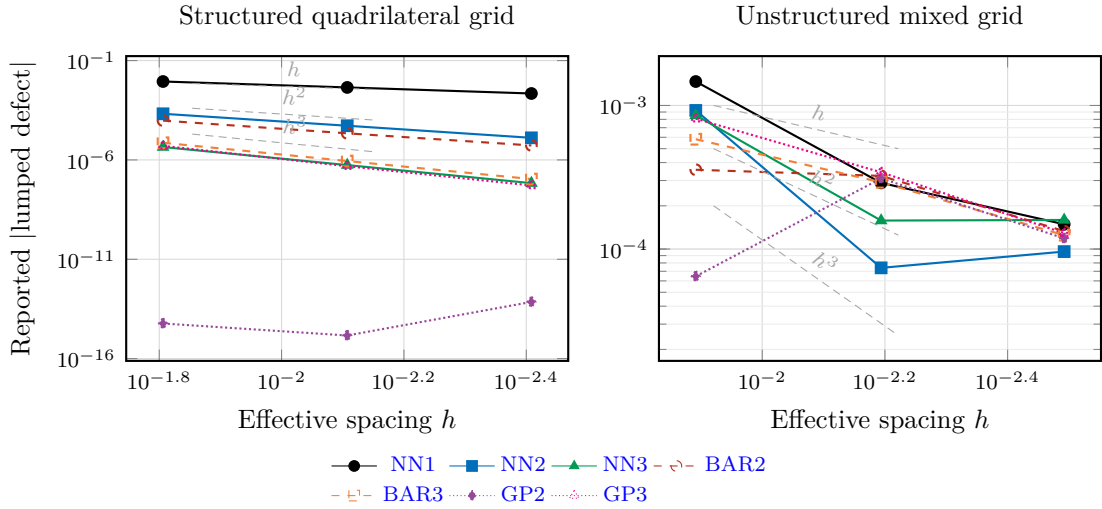


Figure 10: Solver-lumped density-integral defect for the discontinuous transfer before global correction. The gray $O(h)$, $O(h^2)$, and $O(h^3)$ segments are visual guides; no single asymptotic rate is assumed for this discontinuous problem.

smooth mesh, GP2 has a defect near 4.3×10^{-12} , compared with 2.5×10^{-7} on the mixed grid. This is consistent with different weight compatibility, but the archive does not include the basis-integral audit needed to establish that explanation uniquely.

The Riemann-family smooth check uses $\rho = 1 + 0.5 \sin(2\pi x) \cos(2\pi y)$, four measured spacings 0.07752, 0.03984, 0.01996, and 0.01, and mean displacement/spacing approximately 0.057. Figure 11 and Table 6 use the supplied area-weighted L_1 errors over moved nodes. NN2 approaches its second-order slope from above. GP3 approaches three from below, while BAR3 approaches three from above. The correction comparison uses the finest mesh and is not a separate convergence study.

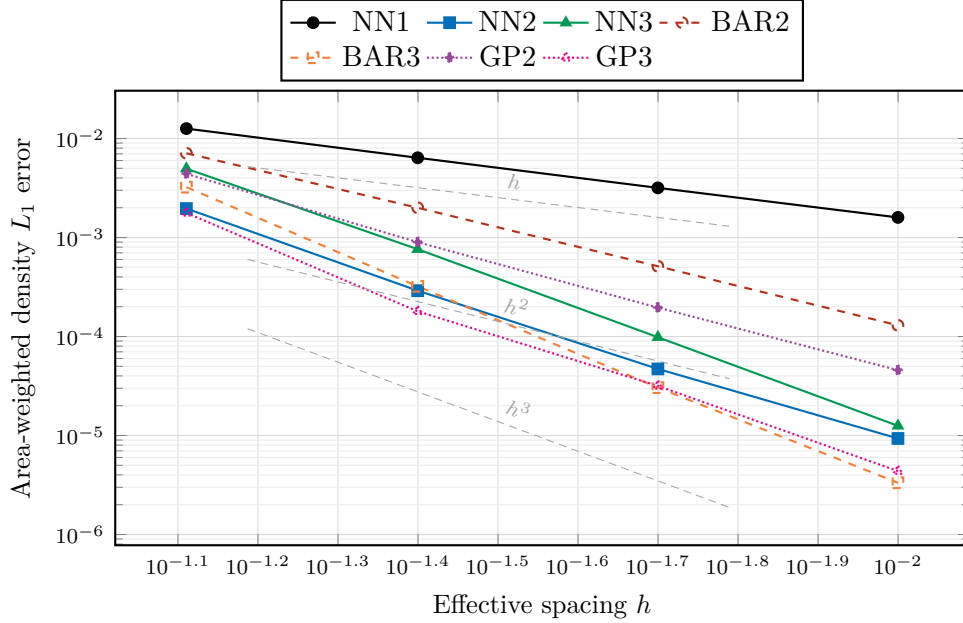


Figure 11: Density L_1 transfer error on the structured Riemann mesh family, weighted by nodal control-volume area. Gray guides show $O(h)$, $O(h^2)$, and $O(h^3)$ behavior.

Table 6: Observed density L_1 convergence orders on the structured Riemann mesh family. The fitted order uses all four meshes; the remaining columns give the order between consecutive mesh levels.

Scheme	p_{nom}	p_{fit}	$p_{1 \rightarrow 2}$	$p_{2 \rightarrow 3}$	$p_{3 \rightarrow 4}$
NN1	1	1.01	1.02	1.01	1
NN2	2	2.61	2.87	2.63	2.34
NN3	3	2.93	2.82	2.96	2.98
BAR2	2	1.96	1.9	1.97	1.99
BAR3	3	3.36	3.46	3.41	3.2
GP2	2	2.23	2.39	2.2	2.11
GP3	3	2.89	3.44	2.51	2.87

Table 8: Per-transfer density conservation and boundedness over the 80-transfer Riemann run. The overshoot is the largest excursion beyond the source range, normalized by that range.

Scheme	Median $ \Delta I_\rho /I_\rho$	Cycles with overshoot	Maximum excursion
BAR2	$8.520 \cdot 10^{-7}$	0	0
BAR2	$4.870 \cdot 10^{-7}$	0	0
BAR3	$4.420 \cdot 10^{-7}$	394	0.02
BAR2	$8.530 \cdot 10^{-7}$	0	0
BAR3	$4.530 \cdot 10^{-7}$	80	0.0082
GP2	$4.940 \cdot 10^{-7}$	398	0.002
GP2	$1.230 \cdot 10^{-7}$	397	0.002
GP3	$4.600 \cdot 10^{-7}$	385	0.002
GP2	$1.040 \cdot 10^{-7}$	80	0.002
GP3	$4.220 \cdot 10^{-7}$	80	0.002
NN1	$1.260 \cdot 10^{-5}$	0	0
NN1	$1.500 \cdot 10^{-15}$	0	0
NN2	$1.500 \cdot 10^{-15}$	389	0.072
NN3	$1.310 \cdot 10^{-15}$	389	0.041
NN1	$2.860 \cdot 10^{-5}$	0	0
NN1	$1.500 \cdot 10^{-15}$	0	0
NN2	$2.460 \cdot 10^{-7}$	389	0.071
NN3	$3.620 \cdot 10^{-7}$	389	0.041
NN1	$8.500 \cdot 10^{-6}$	0	0
NN2	$2.100 \cdot 10^{-7}$	80	0.043
NN3	$4.550 \cdot 10^{-7}$	80	0.026

Table 9: Position and thickness of the density transition at the end of the 80-transfer Riemann run, using the maximum-gradient definition. Position errors are expressed in baseline cells ($h = 0.01$); a positive thickness change denotes additional smearing relative to the no-remeshing reference.

Scheme	x_{tr}	Position error (cells)	Thickness change (cells)
NN1	2.7025	0.14	0.57
NN2	2.7009	-0.02	-0.01
NN3	2.7012	0.01	-0.01
BAR2	2.7012	0.01	0.25
BAR3	2.7011	0	0.01
GP2	2.7012	0.01	0
GP3	2.701	-0.01	0

Table 10: Effect of increasing the remeshing frequency from 80 to 400 transfers at SafeFactor 4. Position and width use the 50%-crossing diagnostic; the density column is the signed sum of the integral changes introduced at the transfers themselves.

Scheme	Cycles	Position error (cells)	Width change (cells)	Cumulative ΔI_ρ
NN1	80	0.14	0.57	$1.10 \cdot 10^{-4}$
NN1	400	0.1	21.41	$5.00 \cdot 10^{-4}$
NN2	80	-0.02	-0.01	$-8.80 \cdot 10^{-6}$
NN2	400	-0.04	0	$-1.70 \cdot 10^{-5}$
NN3	80	0.01	-0.01	$2.20 \cdot 10^{-5}$
NN3	400	0.02	0	$2.70 \cdot 10^{-5}$
BAR2	80	0.01	0.25	$2.00 \cdot 10^{-5}$
BAR2	400	-0.03	3.24	$1.40 \cdot 10^{-4}$
BAR3	80	0	0.01	$-1.70 \cdot 10^{-5}$
BAR3	400	0	0.07	$-5.20 \cdot 10^{-6}$
GP2	80	0.01	0	$2.60 \cdot 10^{-6}$
GP2	400	0.01	0	$-3.00 \cdot 10^{-6}$
GP3	80	-0.01	0	$-2.10 \cdot 10^{-5}$
GP3	400	-0.01	0.02	$1.80 \cdot 10^{-5}$

Table 11: Effect of deformation amplitude at 400 transfers, together with the NN1 global-correction comparison. SafeFactor 2 doubles the displacement bound relative to SafeFactor 4; transition position and width use the same 50%-crossing diagnostic.

Scheme	Global correction	SafeFactor	Cycles	Position error (cells)	Width change (cells)	Cumulative ΔI_ρ
NN1	off	4	400	0.1	21.41	$5.00 \cdot 10^{-4}$
NN1	off	2	400	-0.38	26.36	$1.30 \cdot 10^{-3}$
NN1	on	4	400	-0.15	18.36	$5.80 \cdot 10^{-14}$
NN1	on	2	400	-0.91	25.65	$4.30 \cdot 10^{-14}$
GP2	intrinsic	4	400	0.01	0	$-3.00 \cdot 10^{-6}$
GP2	intrinsic	2	400	0.17	0.06	$-1.10 \cdot 10^{-5}$
BAR2	off	4	400	-0.03	3.24	$1.40 \cdot 10^{-4}$
BAR2	off	2	400	-0.06	22.91	$2.70 \cdot 10^{-4}$

C Riemann moment and limiter diagnostics

The reported lumped moments are $M_0 = \sum_i w_i \rho_i$, $M_1 = \sum_i w_i x_i \rho_i$, and $x_c = M_1/M_0$. They differ from exactly integrated moments of a continuous finite-element field. Their relative errors of 10^{-7} – 10^{-5} are not machine precision. Preservation of M_0 and M_1 does not uniquely determine a transition crossing or its width. GP3 would also preserve consistent quadratic moments before post-processing when their test functions are admissible; the absence of a quadratic moment constraint applies to GP2, not to all Galerkin projections.

Table 12: Change in the solver-lumped zeroth and first density moments during one SafeFactor-2 transfer. The centroid shift is $x_{c,t} - x_{c,s}$, where $x_c = M_1/M_0$.

Scheme	$\Delta M_0/M_0$	$\Delta M_1/M_1$	Δx_c
GP2	$2.30 \cdot 10^{-7}$	$4.90 \cdot 10^{-7}$	$3.70 \cdot 10^{-7}$
NN1	$-2.50 \cdot 10^{-5}$	$-3.90 \cdot 10^{-5}$	$-2.00 \cdot 10^{-5}$
BAR2	$7.50 \cdot 10^{-7}$	$3.10 \cdot 10^{-6}$	$3.40 \cdot 10^{-6}$

Table 13: GP2 limiter-activation diagnostic for a single transfer. The table reports the range violation of the unlimited projection before clipping and conservative redistribution, allowing the two deformation amplitudes to be compared.

SafeFactor	Mean displacement	Unlimited excursion	Fraction near transition
4	$9.73 \cdot 10^{-4}$	$8.20 \cdot 10^{-6}$	—
2	$1.94 \cdot 10^{-3}$	$3.40 \cdot 10^{-3}$	0.967

The activation table omits the source report’s categorical “dormant/active” designation: exceeding a global source range is not the same test as exceeding every local projection bound. The 97% concentration reported at SF2 supports localization of the measured excursion. It does not quantify the portion of accumulated width attributable to repair, and the two available amplitudes do not define a scaling law.

D Additional NACA history and force diagnostics

The following tables preserve the archived values. A density “integral difference” is target-referenced in the transfer and GC tables; it is not a source-referenced conservation error. The full first-step matrix contains five methods, whereas the ratio table contains all seven raw variants. The relaxation file contains first, peak, and final summaries only, despite its history-like filename. It is not used to generate a synthetic time history.

Table 14: Target-referenced discrepancy after transferring the converged Euler NACA 0012 solution from mesh A to mesh B . The norm includes all stored flow variables and is measured against the independently converged state on mesh B .

Method	$\ \varepsilon^*\ _{M_B}$	Relative norm	Density integral difference
NN1	$1.474 \cdot 10^{-3}$	$3.64 \cdot 10^{-6}$	$-7.287 \cdot 10^{-4}$
NN2	$7.237 \cdot 10^{-4}$	$1.79 \cdot 10^{-6}$	$-7.939 \cdot 10^{-4}$
NN3	$7.151 \cdot 10^{-4}$	$1.76 \cdot 10^{-6}$	$-7.985 \cdot 10^{-4}$
BAR2	$7.034 \cdot 10^{-4}$	$1.74 \cdot 10^{-6}$	$-8.138 \cdot 10^{-4}$
BAR3	$7.111 \cdot 10^{-4}$	$1.75 \cdot 10^{-6}$	$-7.913 \cdot 10^{-4}$
GP2	$7.132 \cdot 10^{-4}$	$1.76 \cdot 10^{-6}$	$-8.123 \cdot 10^{-4}$
GP3	$7.120 \cdot 10^{-4}$	$1.76 \cdot 10^{-6}$	$-7.855 \cdot 10^{-4}$

Table 15: First physical-step response of the mixed-history Euler NACA restarts. State and force differences are measured against the native continuation on the target mesh.

Method	Histories	$\ \delta^{n+1}\ _{M_B}$	ΔC_L	ΔC_D	ΔC_m
NN1	full	$1.461 \cdot 10^{-3}$	$5.787 \cdot 10^{-3}$	$1.125 \cdot 10^{-2}$	$1.331 \cdot 10^{-4}$
NN1	latest	$1.948 \cdot 10^{-3}$	$7.714 \cdot 10^{-3}$	$1.502 \cdot 10^{-2}$	$1.772 \cdot 10^{-4}$
NN1	older	$4.872 \cdot 10^{-4}$	$-1.840 \cdot 10^{-3}$	$-3.629 \cdot 10^{-3}$	$-5.163 \cdot 10^{-5}$
NN2	full	$7.224 \cdot 10^{-4}$	$-6.884 \cdot 10^{-3}$	$5.106 \cdot 10^{-3}$	$1.880 \cdot 10^{-3}$
NN2	latest	$9.630 \cdot 10^{-4}$	$-9.168 \cdot 10^{-3}$	$6.808 \cdot 10^{-3}$	$2.499 \cdot 10^{-3}$
NN2	older	$2.413 \cdot 10^{-4}$	$2.313 \cdot 10^{-3}$	$-1.710 \cdot 10^{-3}$	$-6.273 \cdot 10^{-4}$
BAR2	full	$7.026 \cdot 10^{-4}$	$-7.285 \cdot 10^{-3}$	$6.296 \cdot 10^{-3}$	$1.674 \cdot 10^{-3}$
BAR2	latest	$9.367 \cdot 10^{-4}$	$-9.709 \cdot 10^{-3}$	$8.390 \cdot 10^{-3}$	$2.216 \cdot 10^{-3}$
BAR2	older	$2.347 \cdot 10^{-4}$	$2.446 \cdot 10^{-3}$	$-2.106 \cdot 10^{-3}$	$-5.591 \cdot 10^{-4}$
GP2	full	$7.123 \cdot 10^{-4}$	$-7.575 \cdot 10^{-3}$	$6.133 \cdot 10^{-3}$	$1.787 \cdot 10^{-3}$
GP2	latest	$9.494 \cdot 10^{-4}$	$-1.009 \cdot 10^{-2}$	$8.172 \cdot 10^{-3}$	$2.372 \cdot 10^{-3}$
GP2	older	$2.379 \cdot 10^{-4}$	$2.539 \cdot 10^{-3}$	$-2.055 \cdot 10^{-3}$	$-5.998 \cdot 10^{-4}$
GP3	full	$7.110 \cdot 10^{-4}$	$-7.107 \cdot 10^{-3}$	$5.470 \cdot 10^{-3}$	$1.672 \cdot 10^{-3}$
GP3	latest	$9.479 \cdot 10^{-4}$	$-9.471 \cdot 10^{-3}$	$7.291 \cdot 10^{-3}$	$2.218 \cdot 10^{-3}$
GP3	older	$2.375 \cdot 10^{-4}$	$2.387 \cdot 10^{-3}$	$-1.832 \cdot 10^{-3}$	$-5.577 \cdot 10^{-4}$

Table 16: Linearity check for the mixed-history response. The table compares the full restart with the sum of the latest-only and older-only responses and reports the BDF2 response ratios; state norms are positive, whereas the lift ratios retain their sign.

Method	State superposition	State latest/full	State older/full	C_L superposition	C_L latest/full	C_L older/full
NN1	0.0181	1.3333	0.3335	0.0149	1.3329	-0.3179
NN2	0.0353	1.3331	0.3341	0.0042	1.3318	-0.336
NN3	0.034	1.3331	0.334	0.0053	1.3309	-0.3362
BAR2	0.0336	1.3332	0.334	0.0031	1.3328	-0.3358
BAR3	0.0334	1.3332	0.334	0.0028	1.3329	-0.3357
GP2	0.0339	1.333	0.334	0.003	1.3322	-0.3352
GP3	0.0342	1.3331	0.334	0.0033	1.3326	-0.3359

Table 17: Effect of the bounded global conservation correction on the steady Euler NACA restart. The correction preserves the source density integral, so the remaining target-referenced integral difference is the native difference between meshes A and B .

Method	Global	$\ \varepsilon^*\ _{M_B}$	Density integral difference	$\ \delta_{\text{full}}\ _{M_B}$	ΔC_L
NN1	off	$1.474 \cdot 10^{-3}$	$-7.287 \cdot 10^{-4}$	$1.461 \cdot 10^{-3}$	$5.787 \cdot 10^{-3}$
NN1	on	$1.474 \cdot 10^{-3}$	$-8.123 \cdot 10^{-4}$	$1.461 \cdot 10^{-3}$	$5.792 \cdot 10^{-3}$
BAR2	off	$7.034 \cdot 10^{-4}$	$-8.138 \cdot 10^{-4}$	$7.026 \cdot 10^{-4}$	$-7.285 \cdot 10^{-3}$
BAR2	on	$7.034 \cdot 10^{-4}$	$-8.123 \cdot 10^{-4}$	$7.026 \cdot 10^{-4}$	$-7.285 \cdot 10^{-3}$

Table 18: Evolution of the lift discrepancy during the first 50 physical steps after the full-history Euler NACA restart. The table lists the first-step value, the largest excursion in the window, and the final value.

Method	First-step ΔC_L	Maximum $ \Delta C_L $	Reported step	Final ΔC_L
NN1	$5.787 \cdot 10^{-3}$	$2.699 \cdot 10^{-2}$	5	$-1.166 \cdot 10^{-4}$
NN2	$-6.884 \cdot 10^{-3}$	$6.884 \cdot 10^{-3}$	2	$-2.680 \cdot 10^{-3}$
BAR2	$-7.285 \cdot 10^{-3}$	$7.285 \cdot 10^{-3}$	2	$-1.605 \cdot 10^{-3}$
GP2	$-7.575 \cdot 10^{-3}$	$7.575 \cdot 10^{-3}$	2	$-1.923 \cdot 10^{-3}$
GP3	$-7.107 \cdot 10^{-3}$	$7.107 \cdot 10^{-3}$	2	$-2.377 \cdot 10^{-3}$

Table 19: Largest surface-pressure difference produced by the full-history Euler NACA restart on the airfoil nodes shared by the source and target meshes, with the chordwise location of the maximum.

Method	Maximum $ \Delta C_p $	x/c
NN1	0.45	0.001
BAR2	0.4	0.999
GP2	0.29	0.003
GP3	0.3	0.999

Table 20: Check of the exact BDF2 history-residual identity, $\Delta r_{\text{hist}} = -M_B \Delta \eta / \Delta t$. Each comparison uses two history configurations evaluated at the same trial state.

Family	Minimum relative error	Maximum relative error
NN/BAR variants	$1.00 \cdot 10^{-7}$	$1.80 \cdot 10^{-7}$
GP2	$1.30 \cdot 10^{-7}$	$8.80 \cdot 10^{-6}$
GP3	$1.20 \cdot 10^{-7}$	$2.80 \cdot 10^{-6}$

Table 21: Linear trends in lift, drag, and pitching moment over 39 repeated mesh replacements, expressed per 100 physical steps. SE denotes the ordinary regression standard error for each individual time series; it does not account for serial correlation or variability across deformation sequences.

Method	Seed	C_L trend	SE(C_L)	C_D trend	SE(C_D)	C_m trend	SE(C_m)	C_D scatter
GP2	1	$-2.42 \cdot 10^{-6}$	$6.09 \cdot 10^{-6}$	$6.84 \cdot 10^{-6}$	$2.29 \cdot 10^{-6}$	$1.35 \cdot 10^{-7}$	$1.24 \cdot 10^{-6}$	$5.29 \cdot 10^{-5}$
GP2	2	$2.14 \cdot 10^{-6}$	$5.66 \cdot 10^{-6}$	$6.38 \cdot 10^{-6}$	$2.21 \cdot 10^{-6}$	$-1.04 \cdot 10^{-6}$	$1.15 \cdot 10^{-6}$	$5.11 \cdot 10^{-5}$
NN2	1	$1.16 \cdot 10^{-5}$	$7.02 \cdot 10^{-6}$	$3.95 \cdot 10^{-5}$	$3.32 \cdot 10^{-6}$	$-1.46 \cdot 10^{-5}$	$1.53 \cdot 10^{-6}$	$8.85 \cdot 10^{-5}$
NN2	2	$4.87 \cdot 10^{-5}$	$8.39 \cdot 10^{-6}$	$3.84 \cdot 10^{-5}$	$3.53 \cdot 10^{-6}$	$-1.49 \cdot 10^{-5}$	$1.67 \cdot 10^{-6}$	$9.20 \cdot 10^{-5}$

Table 22: Lift-response onset after a single localized mesh replacement. Lags are measured from the transfer event for several fractions of each case’s peak response and are listed together with the freestream acoustic travel-time estimate.

d/c	1%	5%	25%	50%	Peak	$845d$
0.05	16	19	26	31	74	42.3
0.1	38	43	74	80	96	84.5
0.1	37	44	75	82	97	84.5
0.2	105	115	144	171	203	169
0.2	79	92	118	130	199	169

Table 23: Distance-lag fits for the localized Euler NACA disturbance. Force-based slopes use the three launch radii; pressure-probe speeds use different onset thresholds applied to the same event, so the rows measure detector sensitivity rather than independent realizations.

					Pressure threshold (%)	Slope	R^2	Speed
					2	461	0.792	2.17
					5	635	0.92	1.575
					10	753	0.969	1.328
					25	911	0.998	1.097
					50	983	0.999	1.018
					75	1,000	1	1
					2	572	0.933	1.749
					2	929	0.999	1.077
					5	704	0.967	1.421
					5	1,000	1	1
					10	769	0.992	1.3
					10	871	0.999	1.148
					25	908	0.998	1.101
					25	800	1	1.25
					50	832	0.995	1.201
					50	757	0.974	1.321
					75	892	0.94	1.122
					75	-286	0.188	-3.5
					2	528	0.837	1.894
					2	1,300	0.983	0.769
					5	699	0.937	1.431
					5	1,329	1	0.753
					10	786	0.98	1.272
					10	1,000	1	1
					25	914	0.995	1.094
					25	943	0.997	1.061
					50	900	0.997	1.111
					50	843	0.935	1.186
					75	1,003	0.992	0.997
					75	871	0.999	1.148
					2	506	0.822	1.975
					2	881	0.951	1.136
					5	616	0.922	1.622
					5	866	0.974	1.155
					10	724	0.966	1.38
					10	917	0.993	1.091
					25	752	0.988	1.33
					25	826	0.999	1.21
					50	760	0.993	1.316
					50	794	0.997	1.259
					75	745	0.997	1.342
					75	789	0.997	1.267
					2	423	0.792	2.365
					2	760	0.924	1.316
					5	584	0.916	1.712
					5	840	0.974	1.191
					10	659	0.968	1.519
					10	823	0.991	1.215
					25	753	0.984	1.328
					25	858	0.995	1.165
					50	831	0.985	1.203
					50	823	0.968	1.215
					75	869	0.996	1.151
					75	903	0.992	1.107

Force threshold (%)	Slope	Intercept	R^2	Slope/845
1	606	-18	0.989	0.72
5	650	-16.5	0.994	0.77
10	660	-6	0.976	0.78
25	773	-8.5	0.991	0.91
50	927	-13.5	0.999	1.1

E Projection verification and vertex-lumping accuracy

The NACA GP3 assembly used for the reported results passes constant and affine patch tests to 3.9×10^{-11} and 1.1×10^{-10} , respectively. A mesh-specific over-coverage defect identified during verification is documented in the accompanying diagnostic table; it is not evidence of a defect in Galerkin projection itself. In two dimensions, a signed polygon traversal requires signed fan-triangle contributions. Replacing them by absolute areas can overcount self-cancelling portions of the traversal and over-assemble the load vector while leaving the target mass matrix unchanged. This convention is not transferred to disjoint three-dimensional clipped tetrahedra, whose positive measures add. Global bounded repair cannot replace coverage and polynomial-reproduction tests, because it can redistribute a local assembly error into a bounded global bias.

Table 24: Verification of the two-dimensional GP3 common-refinement assembly on the mesh A to mesh B transfer: constant and linear patch tests and the target-cell coverage (the transferred value of the unit field).

Diagnostic	Value
Constant patch maximum error	0.000e+00
Linear patch maximum error	1.150e-10
Maximum target-cell coverage	1.000000
Nodes with error above 1e-6	0
Target nodes	22508

E.1 A quadratic counterexample to an automatic third-order lumped defect

Consider $f(x) = x^2$ on $[0, 1]$. Its exact P^2 projection onto any conforming one-dimensional mesh is itself, so there is no projection error. On a uniform mesh with spacing h , retain only the vertices and use the P^1 nodal integration weights. The resulting composite trapezoidal integral is

$$Q_h(f) = h \left[\frac{f(0) + f(1)}{2} + \sum_{i=1}^{1/h-1} f(ih) \right] = \frac{1}{3} + \frac{h^2}{6}. \quad (30)$$

The difference from the consistent quadratic integral is second order despite exact nodal values and exact projection. The same result holds for $f(x, y) = x^2$ on a uniformly triangulated unit square obtained by splitting Cartesian cells; integrating the vertex-linear field in y leaves the same composite rule in x . Two meshes can consequently have a source-target lumped difference $(h_B^2 - h_A^2)/6$ even when the transferred quadratic is exact. The example is a mathematical check, not an additional CFD measurement. It rules out a general claim that vertex restriction of GP3 necessarily leaves an $O(h^3)$ lumped defect or that repairing that defect automatically retains third-order pointwise accuracy.

F RANS supporting diagnostics and export audit

The RANS views use the archived files in `RANS_PostProcessing/`. Each surface-force and surface-norm file contains 680 exact duplicate rows, and the surface-profile file contains 13,320. Removing duplicates only in derived views leaves 398 unique physical steps per drift case, rather than the 478

Table 25: Linear RANS load trends over the repeated-remeshing window, expressed per 100 physical steps and scaled by 10^6 . Total drag comes from the solver force history; pressure and friction contributions come from deduplicated surface snapshots. SE is the ordinary regression standard error for one time series and does not include serial correlation or between-seed variability.

Closure	Method	Seed	b_D	$SE(b_D)$	b_D^p	b_D^v
SA	GP2	1	-1.750	1.352	-1.341	-0.410
SA	GP2	2	-1.704	1.284	-1.264	-0.441
SA	NN2	1	-1.655	1.339	-1.245	-0.411
SA	NN2	2	-2.197	1.293	-1.746	-0.452
SST	GP2	1	0.103	1.385	0.388	-0.286
SST	GP2	2	0.619	1.368	0.941	-0.323
SST	NN2	1	0.165	1.381	0.451	-0.286
SST	NN2	2	0.180	1.361	0.509	-0.328

rows used in the source surface-fit summary. No conflicting values occur at the same case/step key. Table 25 is recalculated with equal weight per physical step. The original exports remain unchanged.

The integral archive contains 40 latest/older history pairs for each drift run. The pairs written at indices (399, 398) have no subsequent physical-step output in the 2–399 history window. Table 3 therefore uses the 39 earlier pairs. Similarly, each nominal one-event case has an initial and terminal export pair, but only the first is followed by the analyzed response. Counting restart files as physical remeshing events would overstate the exposure. The separate relative-integral column is retained at its exported precision: subtracting rounded inventories of order 10^4 cannot reconstruct defects near 10^{-13} reliably.

The report-derived SST floor statistics in Table 26 include both histories and the terminal export, hence 80 transfer calls rather than 39 physical events. No ω floor activation is reported. SA uses no analogous primitive-variable floor; its reported most negative transferred density-weighted $\tilde{v}$ is of order 10^{-16} . The records locate neither the SST clipped vertices nor their volume-weighted contribution to the final aerodynamic response. They establish that the returned transfer is nonlinear, not which individual operation causes the load change.

Table 26: SST kinetic-energy floor activation during repeated remeshing. The table gives, for each method and deformation sequence, the largest number of vertices raised to $k = 10^{-10}$ in one exported history transfer and the smallest pre-floor value encountered across the 80 transferred histories.

Method	Seed	Maximum raised vertices	Minimum k before clipping
GP2	1	423	$-1.29 \cdot 10^{-7}$
GP2	2	415	$-1.62 \cdot 10^{-7}$
NN2	1	19	$-2.38 \cdot 10^{-7}$
NN2	2	13	$-2.17 \cdot 10^{-7}$

For the fixed-band turbulence diagnostic, the RMS is formed over vertices without volume weighting and divided by the maximum reference field magnitude over the same set. Figure 13 displays this metric for eddy viscosity, avoiding comparison of unlike normalizations for k , ω , and $\tilde{v}$. In particular, the large wall value of ω can make a band-maximum-normalized discrepancy small without bounding its relative value everywhere. The archived surface distributions and endpoint tables provide C_p , streamwise C_f , and y^+ differences; the minimum streamwise C_f near the nose is not used alone to diagnose separation.

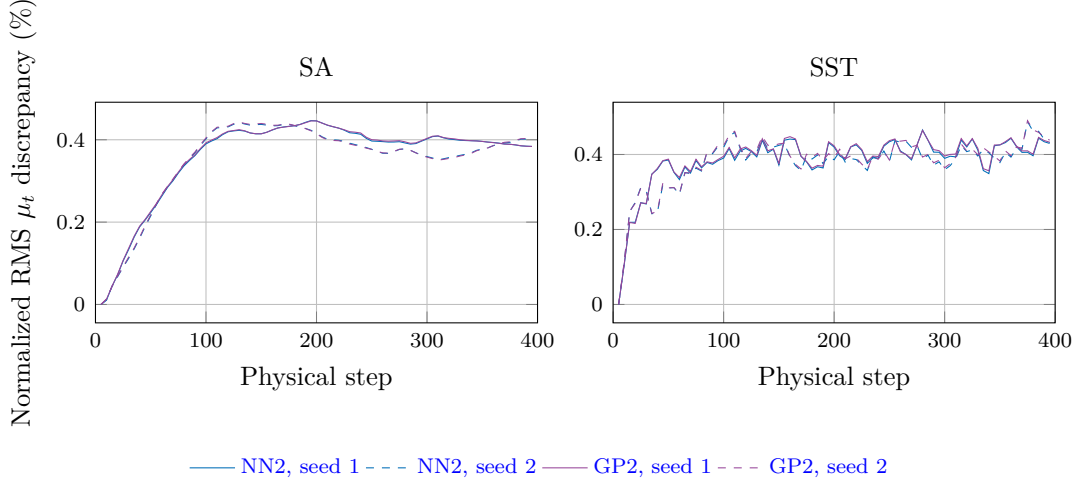


Figure 13: Eddy-viscosity response inside the fixed near-wall vertex set during repeated RANS remeshing. The ordinate is the nodal RMS difference from the no-remeshing reference divided by the maximum reference value on the same 12,233 vertices; markers are available every five physical steps and are connected only to aid readability.

The one-event results are summarized in Table 27. A peak marked C occurs within the last 10% of the available window, so its value and fraction-of-peak onset are window-dependent. The longer $d/c = 0.20$ run begins its event at step 400 rather than 200 and is not simply an extension of the same disturbance: its source state and event-dependent mesh realization also differ. The pressure probes contain 44/72 peaks below the report’s three-increment resolution screen (3×2^{-24} at the exported pressure magnitude). Fitting the remaining zero- or one-step onsets does not yield a resolved acoustic travel-time slope. Accordingly, neither the force-based fits nor the spurious large speeds from three nearly simultaneous shells are used as physical propagation estimates.

Table 27: Lift response after a single localized GP2 remeshing event in the RANS calculations. SF2 doubles the displacement bound. The “long” case uses a later event and a longer observation window; C marks a peak within the last 10% of that window, for which the peak-based onset is sensitive to truncation.

Closure	Case	Peak $ \Delta C_L $	Peak lag	50% lag	1000d/c	
SA	0.05	$2.7 \cdot 10^{-5}$	199	3	50	C
SA	0.10 / SF2	$6.85 \cdot 10^{-5}$	191	9	100	C
SA	0.10	$2.54 \cdot 10^{-5}$	191	8	100	C
SA	0.20 / long	$8.87 \cdot 10^{-6}$	5	3	200	–
SA	0.20	$1.42 \cdot 10^{-5}$	10	5	200	–
SST	0.05	$2.63 \cdot 10^{-5}$	199	3	50	C
SST	0.10 / SF2	$6.75 \cdot 10^{-5}$	191	9	100	C
SST	0.10	$2.53 \cdot 10^{-5}$	191	8	100	C
SST	0.20 / long	$8.31 \cdot 10^{-6}$	5	3	200	–
SST	0.20	$1.38 \cdot 10^{-5}$	10	5	200	–

The RANS archive provides complete reduced force histories and selected surface distributions, but not the case-level meshes, restart vectors, solver sources, or inner-residual histories. Fifty inner iterations is an iteration budget, not evidence that the physical-step error lies below each measured difference. A selected tightened-inner-iteration and time-step check at fixed physical remeshing times is needed before small method differences are interpreted as resolved accuracy differences.

Exact closure variants, freestream turbulence values, thermal wall conditions, the coordinate-based displacement mask, and software version should accompany the configuration archive. These are bounded verification and reporting requirements for this fixed-body study, not a requirement to add moving-body simulations.

G Data provenance and automated checks

The 39 pre-existing input/generated CSVs and 14 RANS source CSVs retain their names, relative locations, and contents. Figures and tables use `pgfplots` and `pgfplotstable`; the build tracks their dependencies. The static builder recomputes convergence orders. The RANS builder creates derived views, rejects conflicting duplicate physical keys, removes only exact repeated snapshots, aligns event windows, and recomputes descriptive statistics. Its hash manifest distinguishes original data from derived products and detects later intentional source updates. Neither builder runs CFD.

The original reports are retained. Earlier Riemann/Euler NACA CSVs contain report-transcribed summaries; the new RANS files contain reduced post-processing exports, including actual sampled force histories. Continuous curves are drawn only where time samples exist. Report-only floor statistics are labeled separately. The earlier residual-summary CSV’s cosines above one remain preserved but excluded from validation. The accompanying scientific assessment records the remaining source/precision and solver-control limitations and the bibliographic additions. Reproduction from these reduced data is not a substitute for archiving the configurations, meshes, restart states, and extraction scripts required to reproduce the simulations themselves.

References

- [1] Frédéric Alauzet, Adrien Loseille, and Géraldine Olivier. “Time-Accurate Multi-Scale Anisotropic Mesh Adaptation for Unsteady Flows in CFD”. In: *Journal of Computational Physics* 373 (2018), pp. 28–63. DOI: [10.1016/j.jcp.2018.06.043](https://doi.org/10.1016/j.jcp.2018.06.043).
- [2] Bastien Sauvage, Frédéric Alauzet, and Alain Dervieux. “A space and time fixed point mesh adaptation method”. In: *Journal of Computational Physics* 519 (2024), p. 113389. DOI: [10.1016/j.jcp.2024.113389](https://doi.org/10.1016/j.jcp.2024.113389).
- [3] Patrick E. Farrell and James R. Maddison. “Conservative Interpolation between Volume Meshes by Local Galerkin Projection”. In: *Computer Methods in Applied Mechanics and Engineering* 200.1–4 (2011), pp. 89–100. DOI: [10.1016/j.cma.2010.07.015](https://doi.org/10.1016/j.cma.2010.07.015).
- [4] Xiangmin Jiao and Michael T. Heath. “Common-Refinement-Based Data Transfer between Non-Matching Meshes in Multiphysics Simulations”. In: *International Journal for Numerical Methods in Engineering* 61 (2004), pp. 2402–2427.
- [5] J. R. Maddison and H. R. Hiester. “Optimal Constrained Interpolation in Mesh-Adaptive Finite Element Modeling”. In: *SIAM Journal on Scientific Computing* 39.5 (2017), A2257–A2286. DOI: [10.1137/15M102054X](https://doi.org/10.1137/15M102054X).
- [6] J. R. Maddison, C. J. Cotter, and P. E. Farrell. “Geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling”. In: *Ocean Modelling* 37.1–2 (2011), pp. 35–48. DOI: [10.1016/j.ocemod.2010.12.007](https://doi.org/10.1016/j.ocemod.2010.12.007).
- [7] J. R. Maddison, C. J. Cotter, and P. E. Farrell. “Corrigendum to “Geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling” [Ocean Model. 37 (2011) 35–48]”. In: *Ocean Modelling* 68 (2013), p. 106. DOI: [10.1016/j.ocemod.2013.04.007](https://doi.org/10.1016/j.ocemod.2013.04.007).

- [8] Timothy J. Barth and Paul O. Frederickson. “Higher Order Solution of the Euler Equations on Unstructured Grids Using Quadratic Reconstruction”. In: *28th Aerospace Sciences Meeting*. AIAA Paper 90-0013. Reno, Nevada, 1990.
- [9] Timothy J. Barth. “Recent Developments in High Order k -Exact Reconstruction on Unstructured Meshes”. In: *31st Aerospace Sciences Meeting*. AIAA Paper 93-0668. Reno, Nevada, 1993.
- [10] Dimitri J. Mavriplis. “Revisiting the Least-Squares Procedure for Gradient Reconstruction on Unstructured Meshes”. In: *16th AIAA Computational Fluid Dynamics Conference*. AIAA Paper 2003-3986. Orlando, Florida, 2003. DOI: [10.2514/6.2003-3986](https://doi.org/10.2514/6.2003-3986).
- [11] Boris Diskin and James L. Thomas. *Accuracy of Gradient Reconstruction on Grids with High Aspect Ratio*. Tech. rep. NIA Report 2008-12. Hampton, Virginia: National Institute of Aerospace, 2008.
- [12] Timothy J. Barth and Dennis C. Jespersen. “The Design and Application of Upwind Schemes on Unstructured Meshes”. In: *27th Aerospace Sciences Meeting*. AIAA Paper 89-0366. Reno, Nevada, 1989. DOI: [10.2514/6.1989-366](https://doi.org/10.2514/6.1989-366).
- [13] V. Venkatakrishnan. “Convergence to Steady State Solutions of the Euler Equations on Unstructured Grids with Limiters”. In: *Journal of Computational Physics* 118.1 (1995), pp. 120–130.
- [14] Kristopher Michalak and Carl Ollivier-Gooch. “Accuracy Preserving Limiter for the High-Order Accurate Solution of the Euler Equations”. In: *Journal of Computational Physics* 228.23 (2009), pp. 8693–8711.
- [15] Hiroaki Nishikawa. “New Unstructured-Grid Limiter Functions”. In: *AIAA SciTech Forum*. AIAA Paper 2022-1374. 2022. DOI: [10.2514/6.2022-1374](https://doi.org/10.2514/6.2022-1374).
- [16] David A. Dunavant. “High Degree Efficient Symmetrical Gaussian Quadrature Rules for the Triangle”. In: *International Journal for Numerical Methods in Engineering* 21.6 (1985), pp. 1129–1148.
- [17] Patrick Keast. “Moderate-Degree Tetrahedral Quadrature Formulas”. In: *Computer Methods in Applied Mechanics and Engineering* 55 (1986), pp. 339–348.
- [18] Mikhail Shashkov and Burton Wendroff. “The Repair Paradigm and Application to Conservation Laws”. In: *Journal of Computational Physics* 198.1 (2004), pp. 265–277.
- [19] Milan Kucharik, Mikhail Shashkov, and Burton Wendroff. “An Efficient Linearity-and-Bound-Preserving Remapping Method”. In: *Journal of Computational Physics* 188.2 (2003), pp. 462–471.
- [20] Bruno Koobus and Charbel Farhat. “Second-Order Time-Accurate and Geometrically Conservative Implicit Schemes for Flow Computations on Unstructured Dynamic Meshes”. In: *Computer Methods in Applied Mechanics and Engineering* 170.1–2 (1999), pp. 103–129. DOI: [10.1016/S0045-7825\(98\)00207-2](https://doi.org/10.1016/S0045-7825(98)00207-2).
- [21] Philippe Geuzaine, Céline Grandmont, and Charbel Farhat. “Design and Analysis of ALE Schemes with Provable Second-Order Time-Accuracy for Inviscid and Viscous Flow Simulations”. In: *Journal of Computational Physics* 191.1 (2003), pp. 206–227. DOI: [10.1016/S0021-9991\(03\)00311-5](https://doi.org/10.1016/S0021-9991(03)00311-5).
- [22] Michael B. Giles and Niles A. Pierce. “An Introduction to the Adjoint Approach to Design”. In: *Flow, Turbulence and Combustion* 65 (2000), pp. 393–415. DOI: [10.1023/A:1011430410075](https://doi.org/10.1023/A:1011430410075).

- [23] Philippe R. Spalart and Steven R. Allmaras. “A One-Equation Turbulence Model for Aerodynamic Flows”. In: *La Recherche Aéronautique* 1 (1994), pp. 5–21.
- [24] Florian R. Menter. “Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications”. In: *AIAA Journal* 32.8 (1994), pp. 1598–1605. DOI: [10.2514/3.12149](https://doi.org/10.2514/3.12149).